ENGLISH

GENERAL INFORMATION

The chain gearmotors to which this instruction booklet refers are designed for operating rolling shutters not balanced with springs or weights. If these products are to be used for different applications, it is recommended to first discuss them with the Manufacturer.
The declaration loses its validity if the gearmotor is modified (e.g. wiring reversal).

SAFETY STANDARDS

- **IMPORTANT! – This manual contains important instructions and warnings regarding safety.** Incorrect installation may cause serious injury. Before commencing work, all sections of the manual must be read carefully. If in any doubt, suspend installation and call the Nice Support Service for clarification.
- **IMPORTANT! – This manual contains important instructions. Keep it for future maintenance work and disposal of the product.**
- **IMPORTANT! – Under the latest European legislation, automatic door and gate installations must be compliant with the standards specified in Directive 2006/42/EC (formerly 98/37/EC) (the Machinery Directive) and the standards EN 12445, EN 12453, EN 12635 and EN 13241-1 in particular, which enable conformity of the automated functionality to be declared. In the light of the above, all work involving installation, connection, testing and maintenance of the product must be carried out exclusively by qualified and competent technicians!**

INSTALLATION RULES

- Before commencing the installation, check if the product is suitable for the desired type of use. If it is not suitable, DO NOT continue with the installation.
- The contents of this manual refer to a standard system.
- **All installation and maintenance work must be carried out with the automation system disconnected from the electricity supply.** If the power disconnection device cannot be seen from where the automation system is positioned, then before starting work a notice must be attached to the disconnection device bearing the words “CAUTION! MAINTENANCE IN PROGRESS”.
- The Control unit must be connected to an electricity supply line equipped with protective earthing.
- Handle the product with care during installation, taking care to avoid crushing, denting or dropping it, or contact with liquids of any kind. Keep the product away from sources of heat and naked flames. Failure to observe the above can damage the product, and increase the risk of danger or malfunction. Should this occur, suspend installation work immediately and contact the Nice Support Service.
- The electrical power line must be fitted with a device that enables complete disconnection of the automation from the mains. The disconnection device must have a gap between contacts that ensures complete disconnection in the conditions of overvoltage category III, in compliance with installation regulations.
- Do not modify any part of the product. Prohibited modifications can only lead to malfunctions. The manufacturer declines all responsibility for damage resulting from unauthorized changes made to the product.
- If the gate or door being automated has a pedestrian gate, then the system must include a control device that will inhibit the operation of the motor when the pedestrian gate is open.
- The product’s packaging material must be disposed of in full compliance with local regulations.

Warnings for use

- The product is not intended for use by persons, including children, with limited physical, sensory or mental capacities, or who lack experience or knowledge, unless supervised or trained in the use of the product by a person responsible for their safety.
- Any children near the automation system must be kept under supervision to ensure that they do not play with it.
- Do not allow children to play with the fixed control devices. Keep remote control devices out of the reach of children.

The instructions shall contain at least the following information:
- the business name and full address of the manufacturer and, where applicable, hisauthorized representative;
- model or type reference of the appliance as marked on the appliance itself, except for the serial number;
- the designation of the appliance together with its explanation in case it is given by acombination of letters and/or numbers.
- the general description of the appliance, when needed due to the complexity of the appliance;
- specific precautions if required during installation, operation, adjusting, user maintenance, cleaning, repairing or moving;
- when needed drawings, diagrams, descriptions and explanations necessary for the safe use and user maintenance of the appliance;
- the possible reasonably foreseeable misuse and, whenever relevant, a warning againstthe effects it may have on the safe use of the appliance.

The words “Original instructions” shall appear on the language version(s) verified by the manufacturer or by the authorized representative.
When a translation of the original instructions has been provided by a person introducing the appliance on the market; the meaning of the sentence “Translation of the original instructions” has to appear in the relevant instructions delivered with the appliance.
The instructions for maintenance/service to be done by specialized personnel, mandated by the manufacturer or the authorized representative may be supplied in only one Community language which the specialized personnel understand.
The instructions shall indicate the type and frequency of inspections and maintenance required for safe operation including the preventive maintenance measures.

GENERAL DANGER WARNINGS AND PREVENTIVE SAFETY

According to the VDE 0113 booklet, the emergency stop devices must also be perfectly efficient regardless of the gearmotor operation mode. Any release of the emergency stop device must absolutely not cause an uncontrolled or undefined restarting.

ASSEMBLY INSTRUCTIONS / FIXING THE MECHANISM

Gearmotor coupling

The gearmotor is to be coupled on the end of the winding shaft that must be previously greased to make these tool-less assembly operations easier. If the seat of the key on the shaft is through, the key (tab) must be blocked to prevent accidental movement.

The winding shaft must not be axially secured to the gearmotor (gearmotor side = mobile bearing). Fixing against an axial traverse of the winding shaft is executed on the side opposite the ball bearing using an adjustment screw or a fixing ring **(Fig. 1)**

The fixing brackets enable assembly both horizontally (vertical gearmotor) and vertically (suspended motor). For different assembly positions and for different reduction unit fixing systems (different heights of the fixing brackets and accessories of the rolling shutter systems), contact the manufacturer. The fixing brackets are mounted so that the winding shaft is horizontal.

It is important to ensure that the welded shafts are concentric and aligned with the central axis of the main winding shaft since otherwise the unbalances and angle defects that occur in this case could cause damages or excessive wear to the transmission components.

If an additional coat of paint is applied on the gearmotor, carefully avoid soiling the ring seals (oil seals) of the shaft. The key (tab) is to be blocked on the through seat of the shaft with 1 screw (or, as an alternative, with a fixing ring) to prevent accidental movement. **(Fig. 2)**

PARACHUTE DEVICE BUILT INTO THE REDUCTION UNIT

The gearmotors are positive clutch startup controls with built-in parachute device, irrespective of the rpm and position.
The parachute device follows the movement without load and wear, and it releases in the case of failure of the mechanism.
No electrical disconnection is necessary because in the case of failure of the mechanism, the transmission between motor, hollow shaft and worm screw is interrupted.
After the release of the parachute device, the gearmotor is no longer able to operate and must be replaced.

The parachute device has the following characteristics:
- protection against the breaking and wear of the toothed wheel
- it is independent from the rpm
- it is independent from the direction of rotation
- it is independent from the position
- it is independent from the vibrations
- it does not require maintenance
- it has good damping capacity in the case of intervention

WARNING:
THE BUILT-IN PARACHUTE DEVICE DOES NOT PROTECT AGAINST THE FORCES THAT INFLUENCE THE SYSTEM FROM THE OUTSIDE

EMERGENCY MANUAL RELEASES

The manual emergency release is envisaged for opening and closing the door in case of power failure, so it is to be used only in emergency situations.

- Avoid regular use!!

Warning! Danger of injury in the case of incorrect use!

- Deactivate the main switch before using the manual emergency release.
- The manual emergency release is to be executed only with the motor stopped.
- The manual emergency release manoeuvre is to be executed from a safe place.
- With a gearmotor equipped with brake (pressure), the main door must be opened and closed with the brake open (released).
- On doors not balanced with weights, the brake is to be released only for checking purposes with the door in the low position for safety reasons.
- An undesirable brake release must be prevented with an adequate check (measurement) on site.

  The manual emergency release must not move the door beyond the final positions because in this case the main switch would activate.

  Therefore, operation of the door in electrical mode is no longer possible.

Handle-operated manual emergency release (fig. 4)

- Insert the handle exerting moderate pressure, then turn it up to when it clicks in position. In this way the control voltage is interrupted and the door can no longer be activated electrically.
- Open and/or close the door turning the handle
- The control voltage is reset and the door can again be activated electrically by removing the handle.

Chain version (fig. 5a-d)

- Grasp the red handle / manual control (1) and softly pull until stopping
- to interrupt the control voltage, and therefore prevent activation of the door in electric mode.
- Open and/or close the door with the emergency release chain (2)
- Grasp the green handle /motor control (3) and softly pull until stopping to restore the control voltage, and therefore once again consent operation of the door in electric mode.

  Average duration: 350 cycles!

Modification of the length of the emergency release chain (fig. 6)

- The emergency release chain opens in the connection point and therefore can be extended or shortened by adding or removing links.
- The links must be carefully bent.
- When the emergency release chain length is modified, it is necessary to pay attention not to twist it.

LIMIT SWITCH SETTING

The door must be balanced in all positions when assembling the sectional main door and balancing with the weights according to the procedures prescribed by the standards. Correct balancing is ascertained by manually opening and closing the door with identical activation force in both directions.

1. Mechanical limit switches

The upper and lower deactivation positions of the door are defined by setting the limit switch.

The gearmotor must be electrically connected for making this setting.

To access the limit switch (**Fig. 8a**: limit switch plate with 7 switches - **Fig. 8b** with 8 switches) unscrew the limit switch guard. It is possible to move the door in dead man mode using the built-in "OPEN", "CLOSE" and STOP buttons if the external control devices have not yet been connected with the control station included in the supply.

Switch off the power and reverse the two L1 and L2 phases of the gearmotor if the door does not open when pressing the "OPEN" button.

The door must open even if the gearmotor has been installed upside down when the "OPEN" button is pressed. Otherwise, switch off the power and reverse the two L1 and L2 phases.

It is also necessary to properly correct the two emergency stop limit switches so that they trip after the limit switch.

Switching off with door in low position

The limit switch for switching off with door in low position is to be set as follows (**Fig. 7-8)**:

Move the door to the desired CLOSE position.

Set the contactor cam 3 E↓white) to activate the limit switch.

Tighten the fixing screw **A**.
For accurate adjustment, turn the **screw B**.

Move the door into the desired OPEN position
Set the contactor cam **1 E**↑(green) in order to activate the limit switch.
Tighten fixing screw A.

For accurate adjustment, turn the **screw B**.

The safety limit switches **2 SE**↓ and **4 SE**↑ (red) must be set in order to be released right after the overtaking of the control limit switch.

The safety limit switches **2 SE**↓ and **4 SE**↑ (red) are factory-set in order to follow the limit switch at a short distance.

Check that the fixing screws are in the correct position after the operation test.

The additional limit switches **8 P2**↓ and **7 P2**↑ are closing contacts with zero potential, while the additional limit switches **6 P1**↓ and **5 P1**↑ are switching contacts with zero potential.

In **automatic mode** limit switch 6 is used as a preliminary limit switch. Therefore it is to be set so that it releases when the door reaches a distance of 5 cm from the ground.
In **dead man mode** it is not necessary to set it and it is used as a contact with zero potential.

2. Electronic limit switch (transducer absolute value) (fig. 12-15)

The EES electronic limit switch is a positioning switch with absolute value for rolling shutters, doors and main doors. The assessment, or setting, of the final positions is carried out with gearmotors for doors tuned to the EES.

During assembly simply insert the six-pole plug.

Particular mechanical settings and positionings are not needed.

The terminals for the safety chain (safety switch) are respectively located on the EES side (**A type**) and on the plate beneath the EES (**B type**). (**Fig. 14 and 15**)

See the EES wiring diagram in the gearmotor instruction manual for the UST1K doors.

STAR/DELTA VOLTAGE CONNECTION

Warning! Danger of death due to electric shock.

Before starting the assembly operations, exclude the voltage from the conductors and check that they are really equal to zero.

The commutability of the motor voltage makes it possible to use the gearmotor on both a 3 x 400 V and 3 x 230 V power grid.
The motor is wired in the factory with star connection for a 3 x 400 V power grid.
For connection to a 230 V power grid, switch the motor onto a delta connection.

To commute the voltage onto the motor, wire the ends of the windings as illustrated in **fig. 9-11**.

When fixing the motor cables, pay attention that the cables are long enough to consent solid fixing, in order to create a connection able to remain secure over time.
Pull the conductors to check if the connection is secure.

It is necessary to also adjust the gearmotor (UST1K) by commuting the motor for connection to a 3 x 400 V power grid. To this regard, see the UST1K.

ANNUAL CHECK

BGR 232 directive "**Windows, doors and main doors** with mechanical operation" is applied. It can be obtained from the National Insurance Institute for Industrial Accidents of your trade association or from us on request.

Maintenance operations on **windows, doors and main doors** with mechanical operation must be performed only by personnel authorised by the company, having the necessary experience and skills (BGR 232).

Information for whoever carries out the checks

Gearmotor:
The mechanism is maintenance-free and it is provided with permanent lubrication. Absolutely no rust must form on the output shaft.

Fixings:
Make sure that the fixing screws are all in the correct position and in perfect condition.
Balancing by weights (e.g. in the sectional main doors):
The door must be balanced in all the positions (see the door assembly instructions) when balancing is done with weights according to the procedures prescribed by the standards.

Brake (if present):
During the annual check perfect brake operation is to be ascertained.

In the case of heavy wear of the brake pads, it is necessary to replace the complete brake. Before replacement, disconnect the cables connected to the electric card.

TECHNICAL SPECIFICATIONS

NOTE: all the technical characteristics indicated refer to a temperature falling beetwen -5°C and +40°C. • Nice reserves the right to change all the modifications to the product it deems necessary at any time, but keeping the functions and intended use unaltered.

Technical data

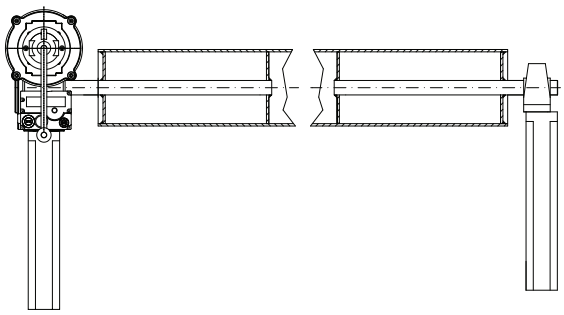
Model **RDF-**	**140-20**	**220-15***	**290-15***	**380-15**	**450-15**	**550-12**	**750-9,5**
Shaft Ø (mm)	30			40			
Max. torque (Nm)	140	220	290	380	450	550	750
Rated torque (Nm)	120	200	250	360	400	450	700
Output rpm	20	15	15	15	15	12	9.5
Motor power (kW)	0.55	1.1	1.1	1.5	1.5	1.5	1.5
Limit switch rpm	15	10-20	10-20	10-20	10-20	10	40
Working voltage (V/Hz)	3x230 V / 3x400 V 50 Hz						
Duty Cycle	S 3 - 60%						
Connection cable (no. x mm^2)	5 x 1.5						
Operating temperature (°C)	-5 ÷ +40						
Noise dB(A)	<70						
Protection rating (IP)	IP 54						
Weight (kg)	12.4	22	24	28.9	27.7	27.7	35

Model **RDF-**	**850-10**	**1100-10**	**1400-7***	**250-24***	**350-24**	**500-24***	**750-24***	**950-24***	**1250-24***	**1400-24***
Shaft Ø (mm)	55			40			55			
Max. torque (Nm)	850	1100	1400	250	350	500	750	950	1250	1400
Rated torque (Nm)	700	950	1250	230	310	450	650	850	1100	1250
Output rpm	10	10	7	24	24	24	24	24	24	24
Motor power (kW)	2.2	2.2	2.2	1.1	1.5	2.2	2.2	3	4	5.5
Limit switch rpm	10-40	10-40	10-40	10-40	10-40	10-40	10-40	10-40	10-40	10-40
Working voltage (V/Hz)	3x230 V / 3x400 V 50 Hz									
Duty Cycle	S 3 - 60%									
Connection cable (no. x mm^2)	5 x 1.5									
Operating temperature (°C)	-5 ÷ +40									
Noise dB(A)	<70									
Protection rating (IP)	IP 54									
Weight (kg)	51.7	51.9	58	24.1	29.2	32.7	51.9	61	63	65

***** *These models are not tested by TÜV-Nord*

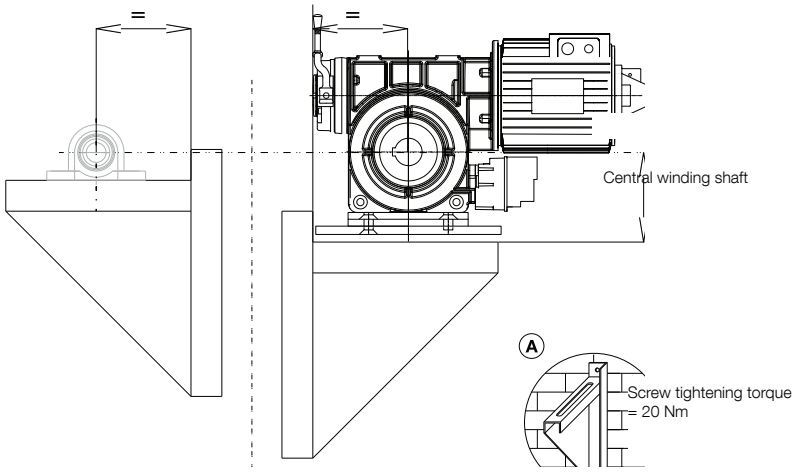
1

Shaft assembly



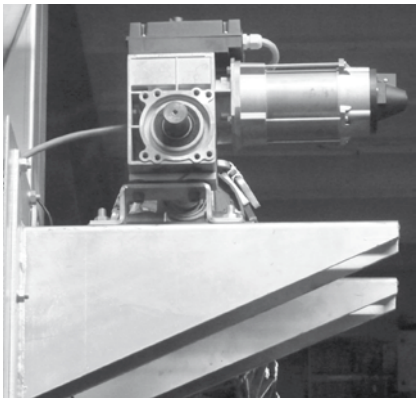
2

Assembly of bracket for large gearmotors

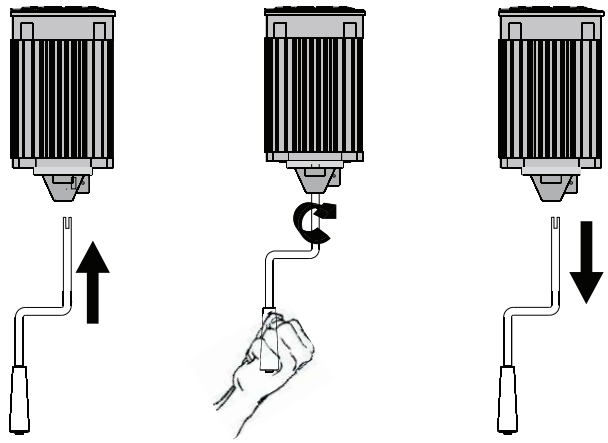


3

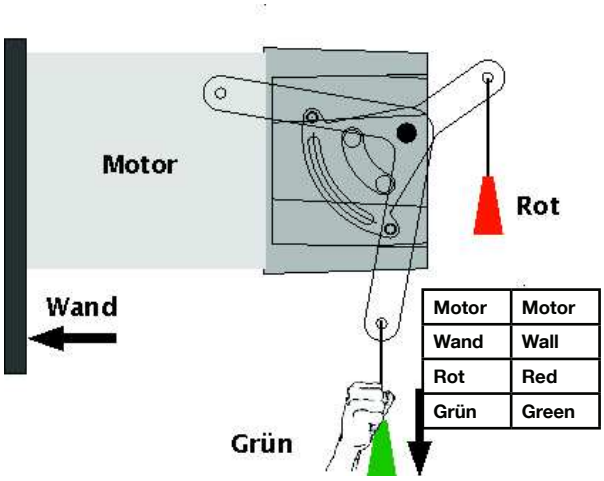
Assembly of bracket for small gearmotors



4

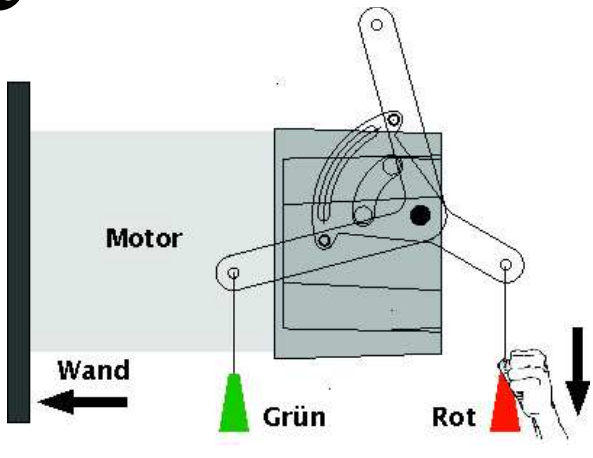


5a

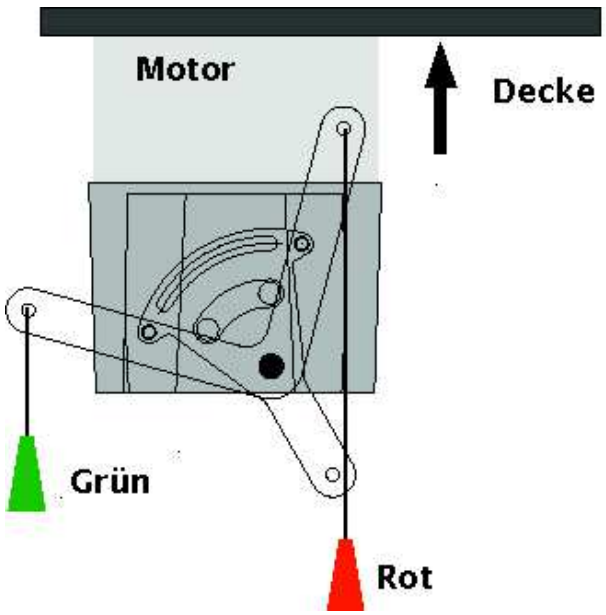


Motor	Motor
Wand	Wall
Rot	Red
Grün	Green

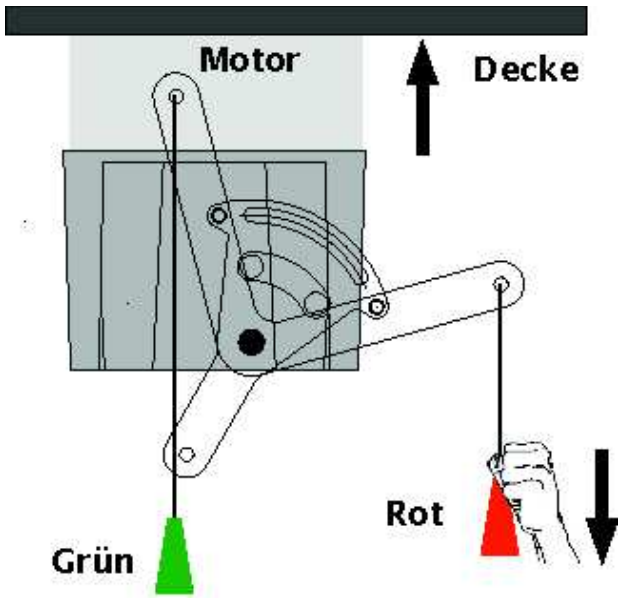
5b



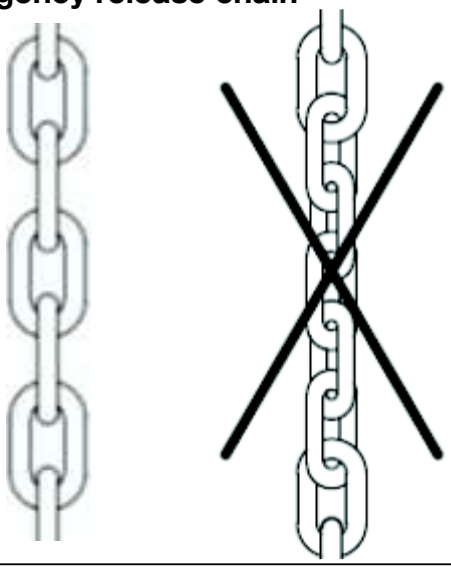
5c



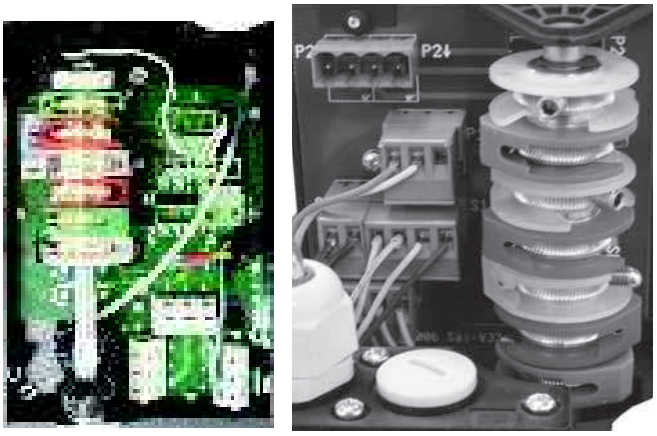
5d



6 Emergency release chain

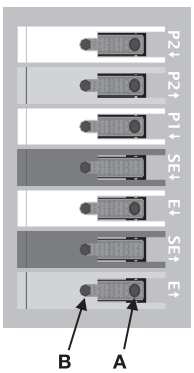


7 Mechanical limit switches



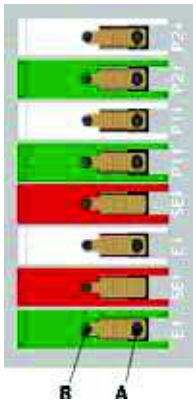
Small types — Large types

8a Mechanical limit switch setting Large types - 7 Contactor cam



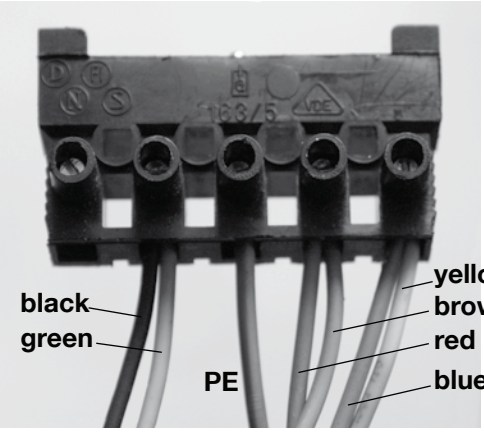
7	white	Additional limit switch 2 CLOSE	Optional
6	green	Additional limit switch 2 OPEN	
5	white	Additional limit switch 1 CLOSE	
4	red	Safety limit switch CLOSE	
3	white	Limit switch CLOSE	
2	red	Safety limit switch OPEN	
1	green	Limit switch OPEN	

8b Mechanical limit switch setting Small itypes - 8 Contactor cam

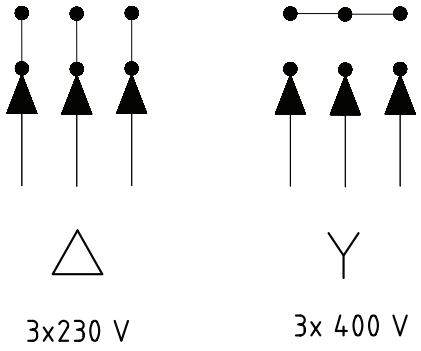


8	white	Additional limit switch 2 CLOSE	Optional
7	green	Additional limit switch 2 OPEN	
6	white	Additional limit switch 1 CLOSE	
5	green	Additional limit switch 1 OPEN	
4	red	Safety limit switch CLOSE	
3	white	Limit switch CLOSE	
2	red	Safety limit switch OPEN	
1	green	Limit switch OPEN	

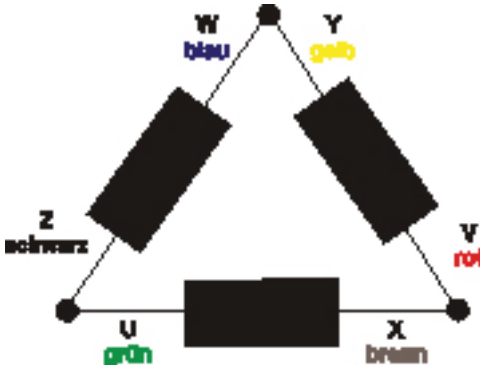
9 Delta connection



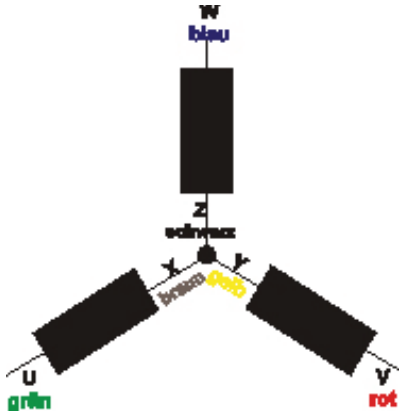
9a Motor connection with terminal block



10 3~230V



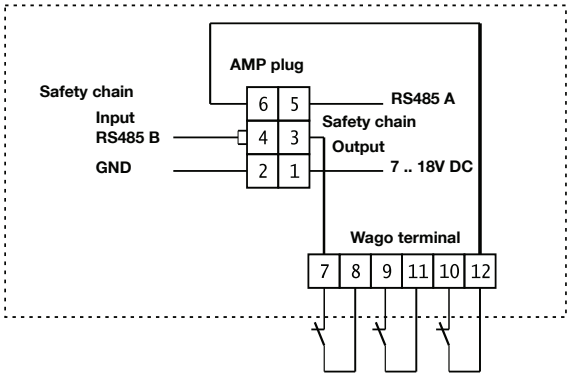
11 Standard wiring 3~400V



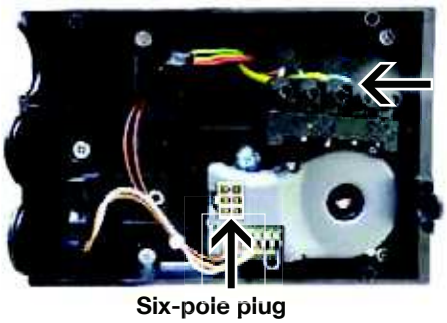
12

Centrale	Reductiemotor	Motor
X2 - J1/U	1	UST1/U
X2 - J1/V	2	UST1/V
X2 - J1/W	3	UST1/W
X2 - J6	Bevestiging draad	-
X5 - J22	4	S3 - J9
X5 - J22	5	S3 - J9
X5 - J21	10	P1↓ - J3/C
X5 - J21	11	P1↓ - J3/NC
X5 - J20	6	E↑ - J2
X5 - J20	7	E↑ - J2
X5 - J19	8	E↓ - J4
X5 - J19	9	E↓ - J4
PE	Geel/groen	PE

13



14 Electronic limit switches
Limit switch housing on the mechanism

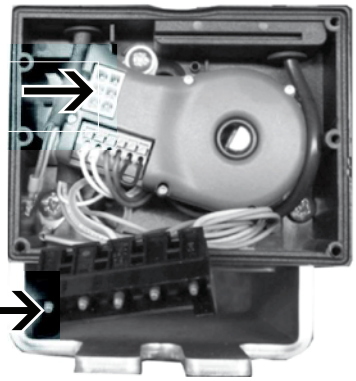


Five-pole multiple adapter Control connection cable

Six-pole plug

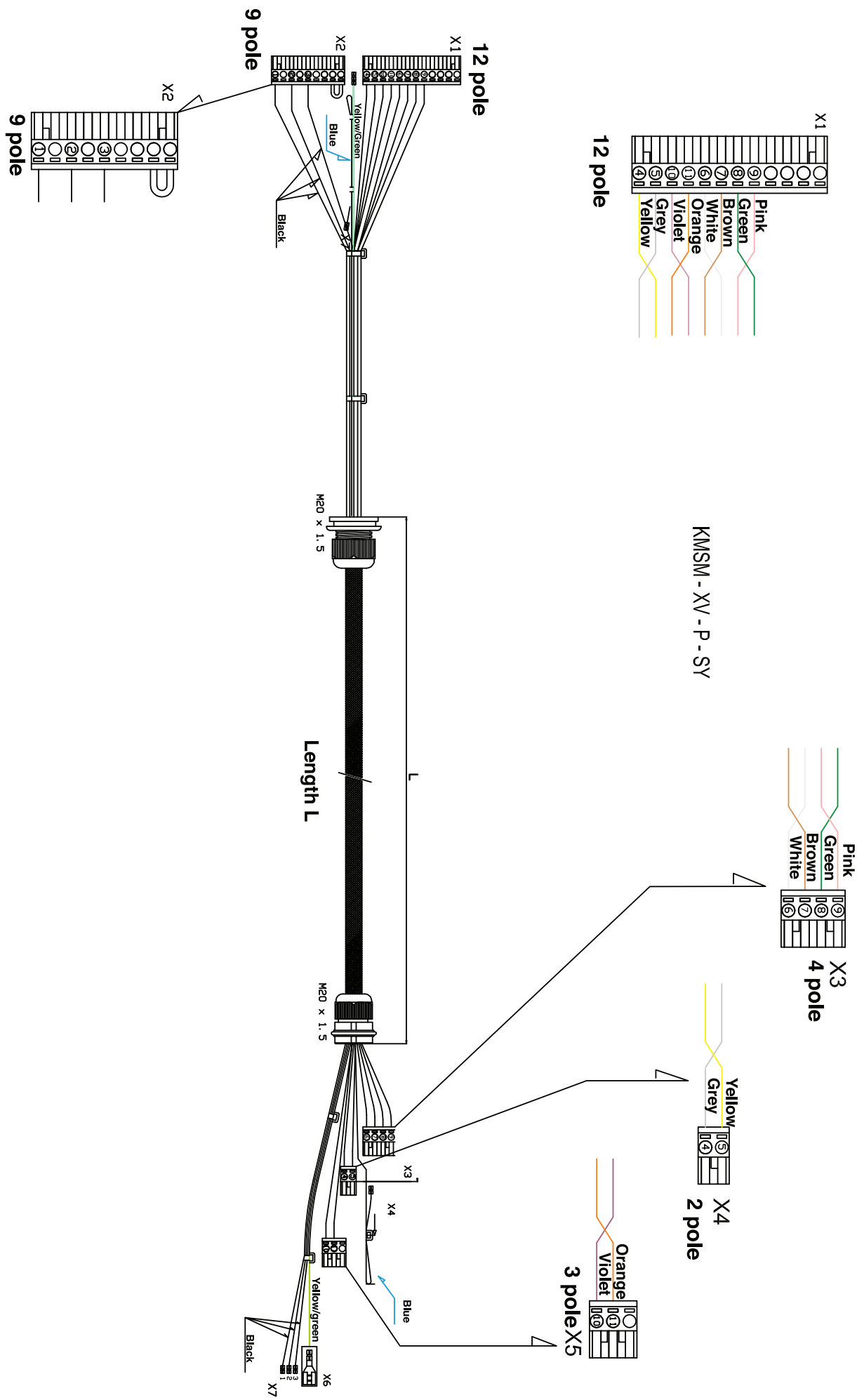
15 Electronic limit switch housing below

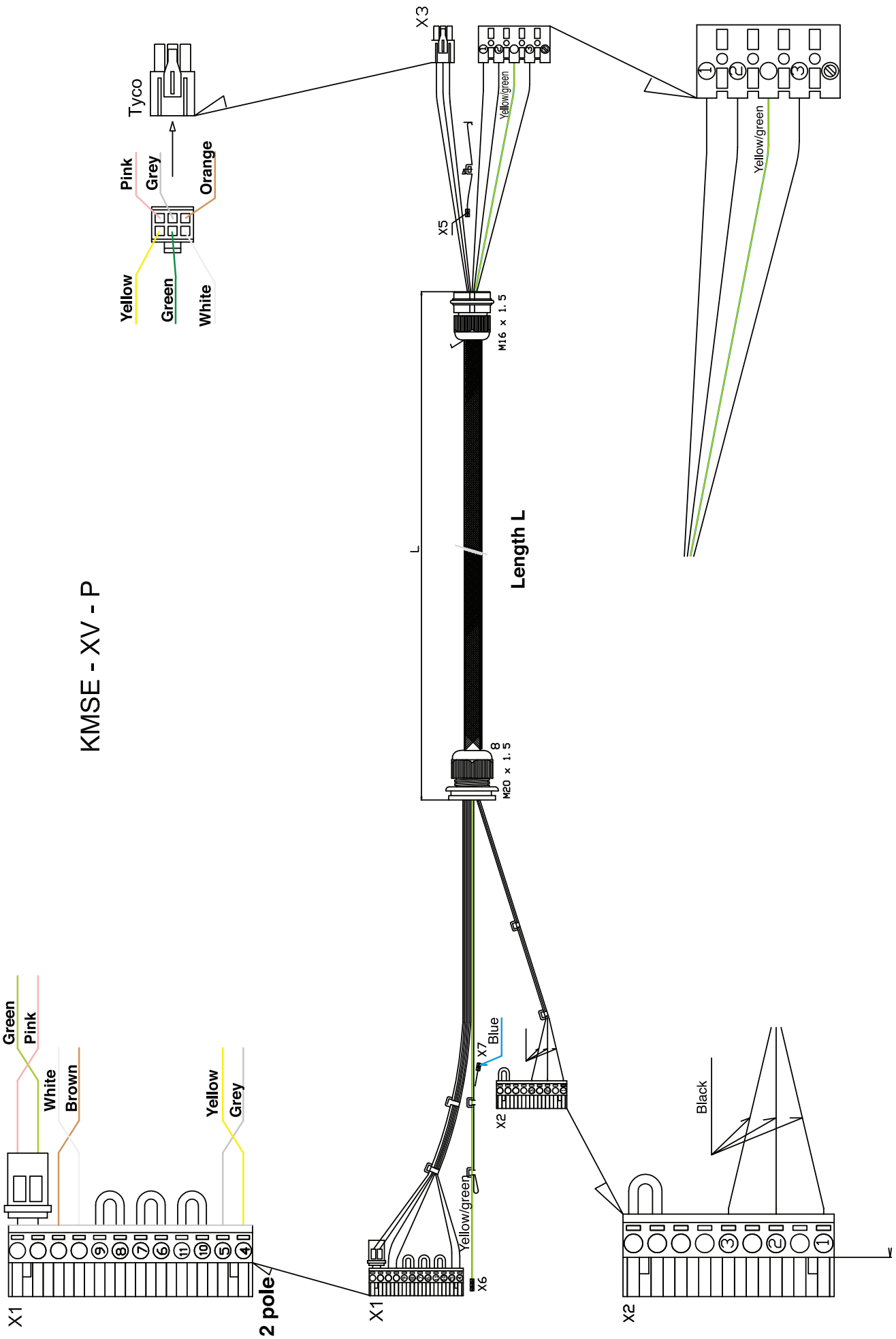
Six-pole plug



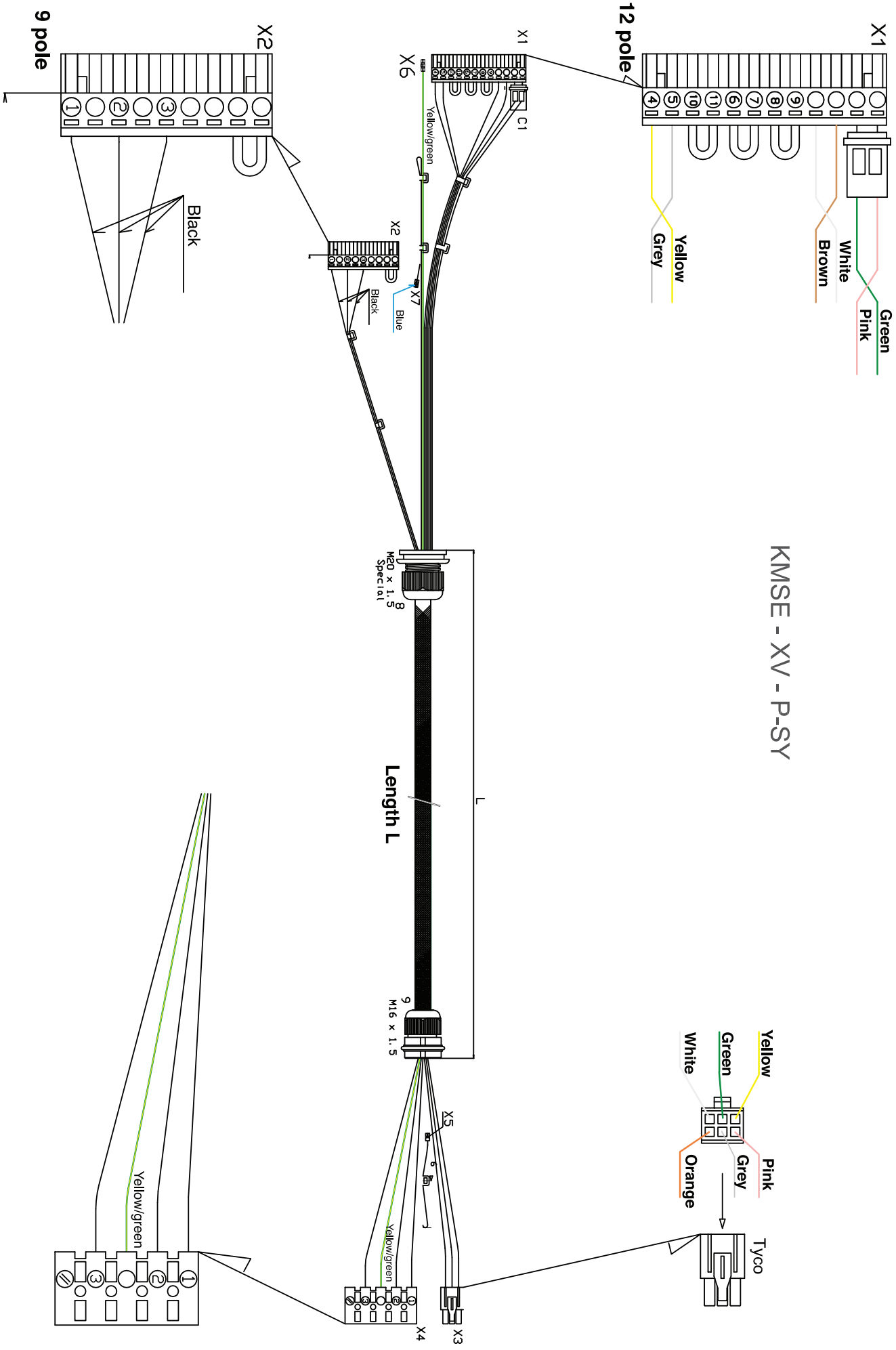
Five-pole multiple adapter Control connection cable

CONNECTION CABLE FOR CONTROLS EQUIPPED WITH MECHANICAL LIMIT SWITCH WITH SCREENING




KMSE - XV - P
Tyco
Pink
Grey
Orange
Yellow
Green
White
X3
X5
Yellow/green
M16 x 1.5
M20 x 1.5
L
Length L
Yellow/green
Green
Pink
White
Brown
Yellow
Grey
12 pole
X1
X7
Blue
X2
Yellow/green
X6
Black
X1
X2

CONNECTION CABLE FOR CONTROLS EQUIPPED WITH ELECTRONIC LIMIT SWITCH WITH SCREENING



Gearmotors with handle and chain

RDF-140-20-KU
RDF-140-20-KE2

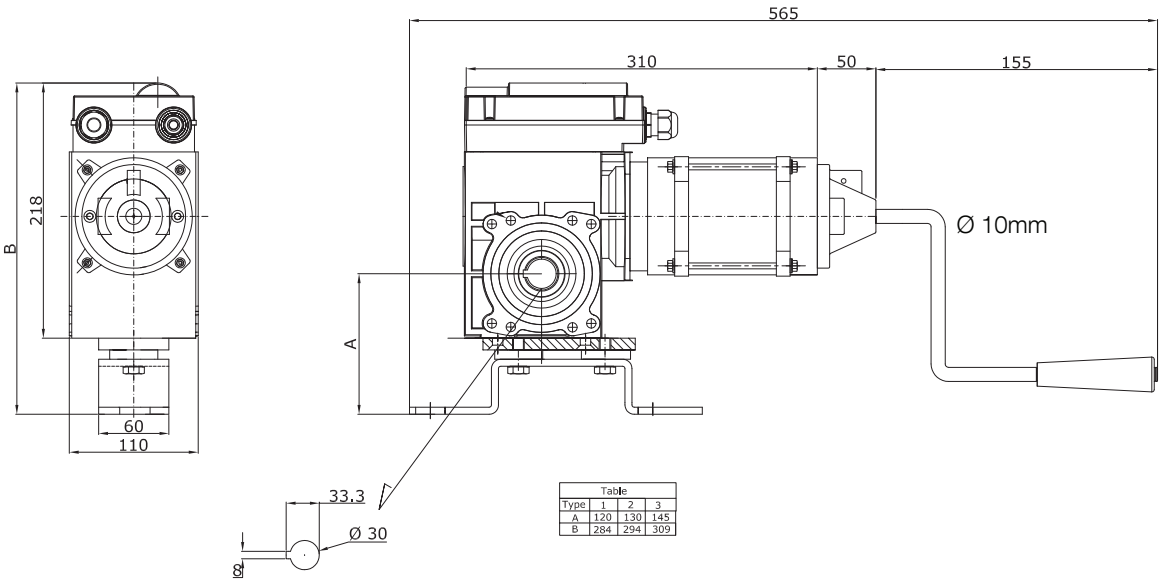


Table			
Type	1	2	3
A	120	130	145
B	284	294	309

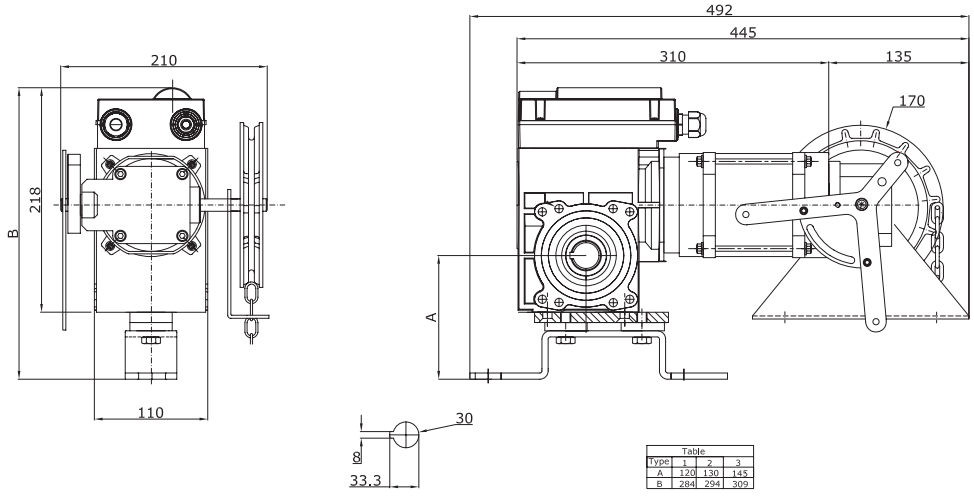


Table			
Type	1	2	3
A	120	130	145
B	284	294	309

Gearmotors with handle and chain

RDF-220/290-15-KU
RDF-220/290-15-KE2

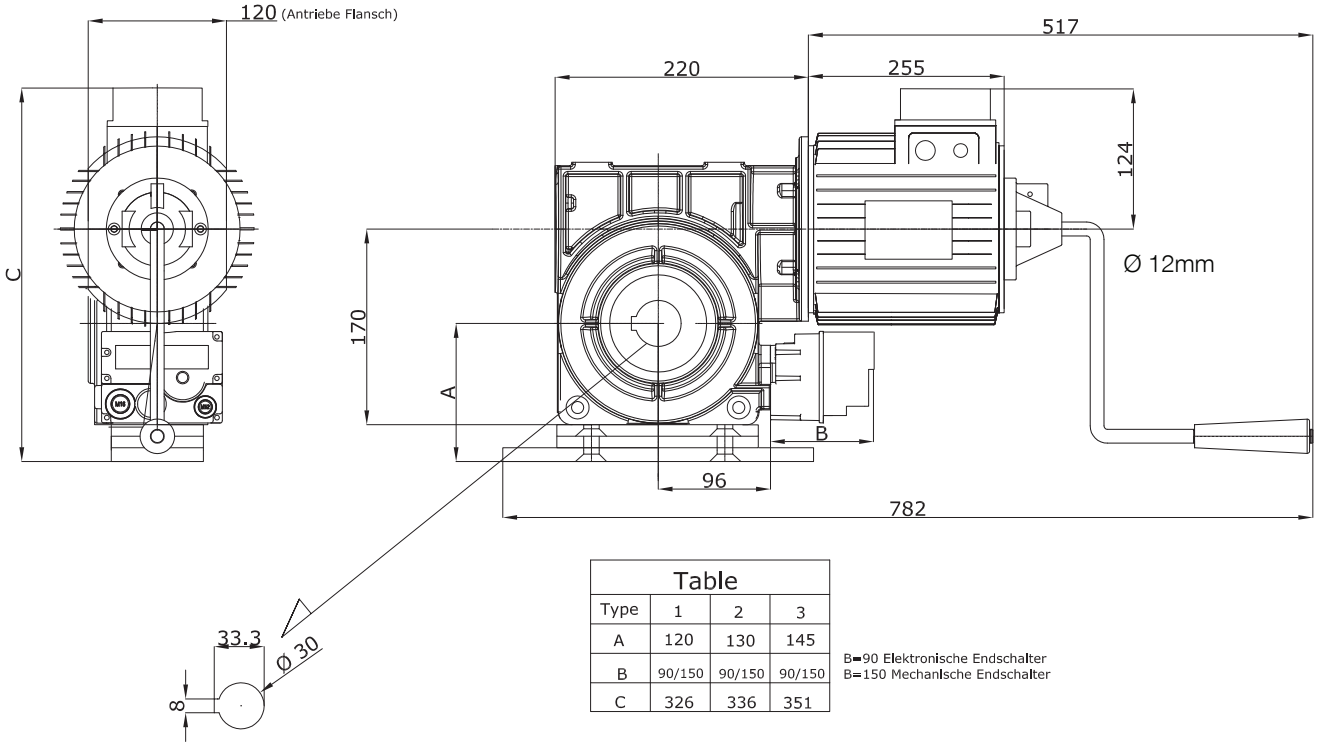
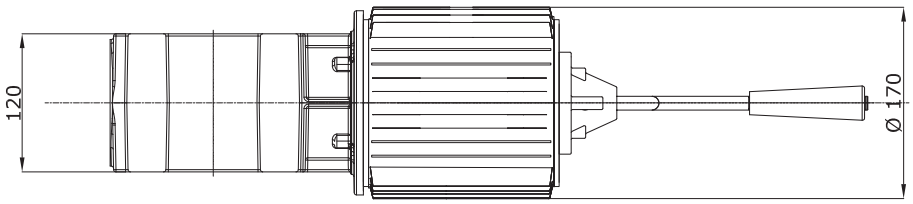


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	326	336	351

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



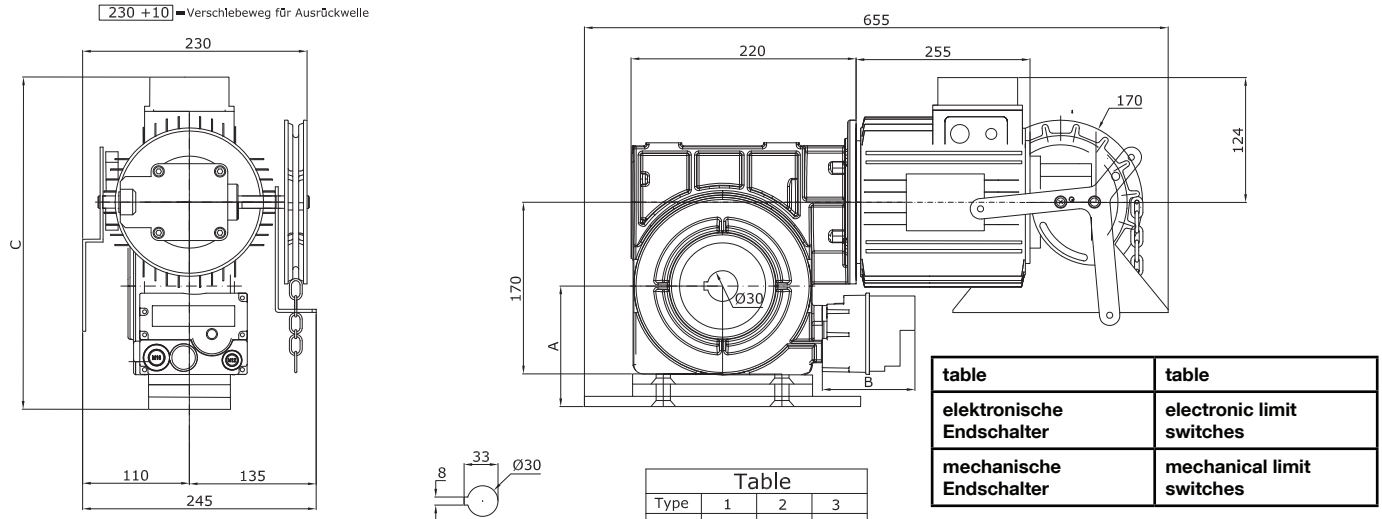
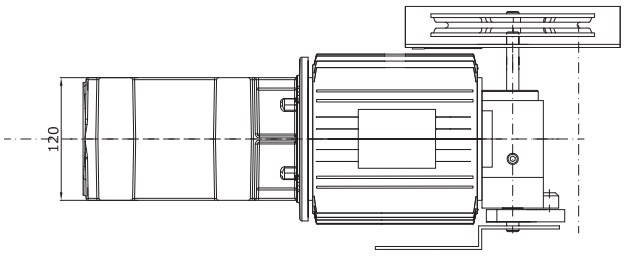


table	table
elektronische Endschalter	electronic limit switches
mechanische Endschalter	mechanical limit switches

Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	326	336	351

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



Gearmotors with handle and chain

RDF-380/450-15-KU
RDF-380/450-15-KE2

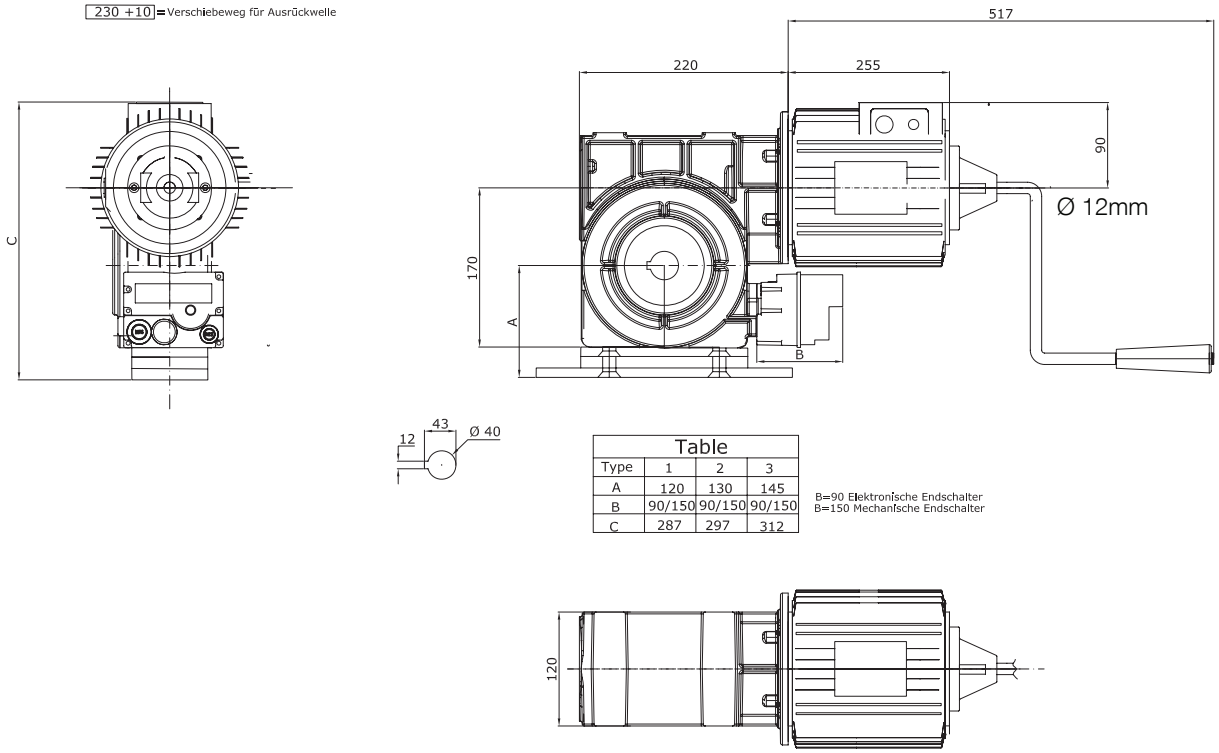


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	287	297	312

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter

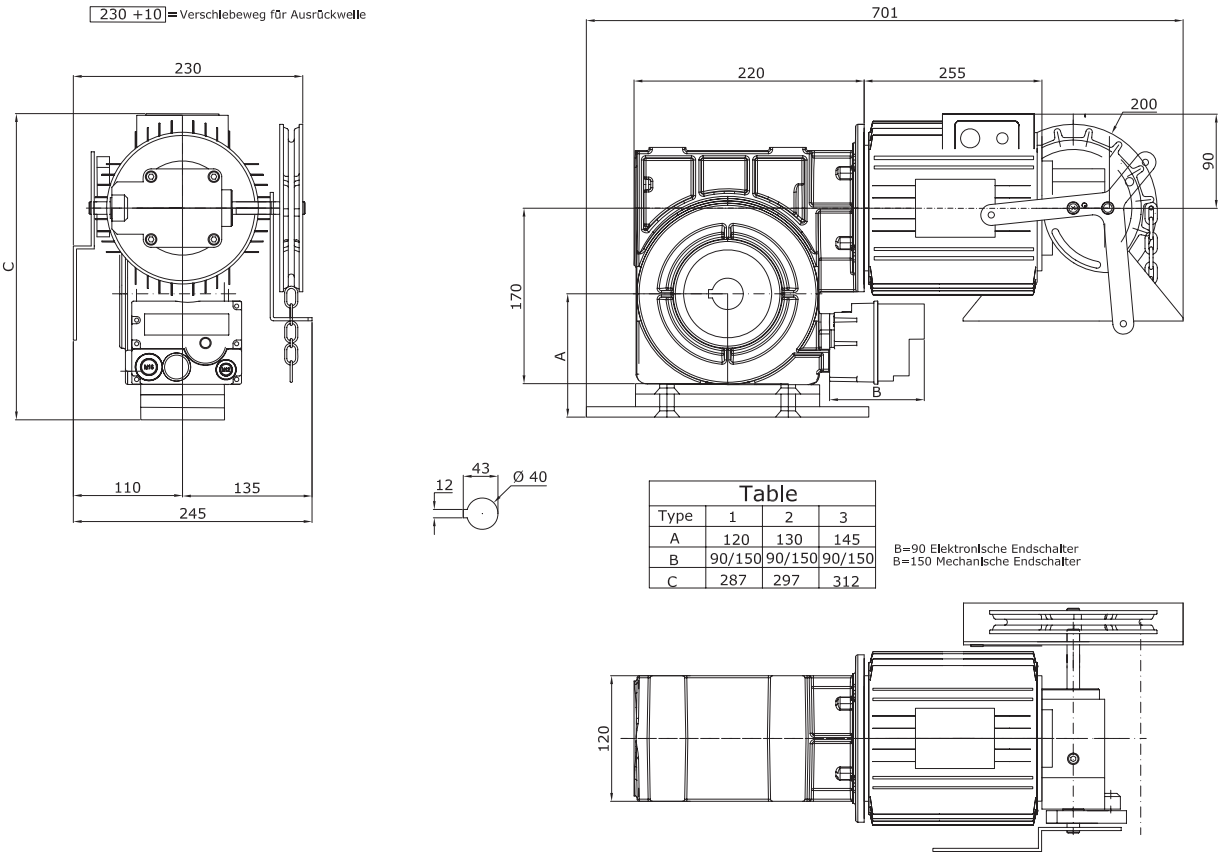


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	287	297	312

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter

Gearmotors with handle and chain

RDF-550-12-KU
RDF-550-12-KE2

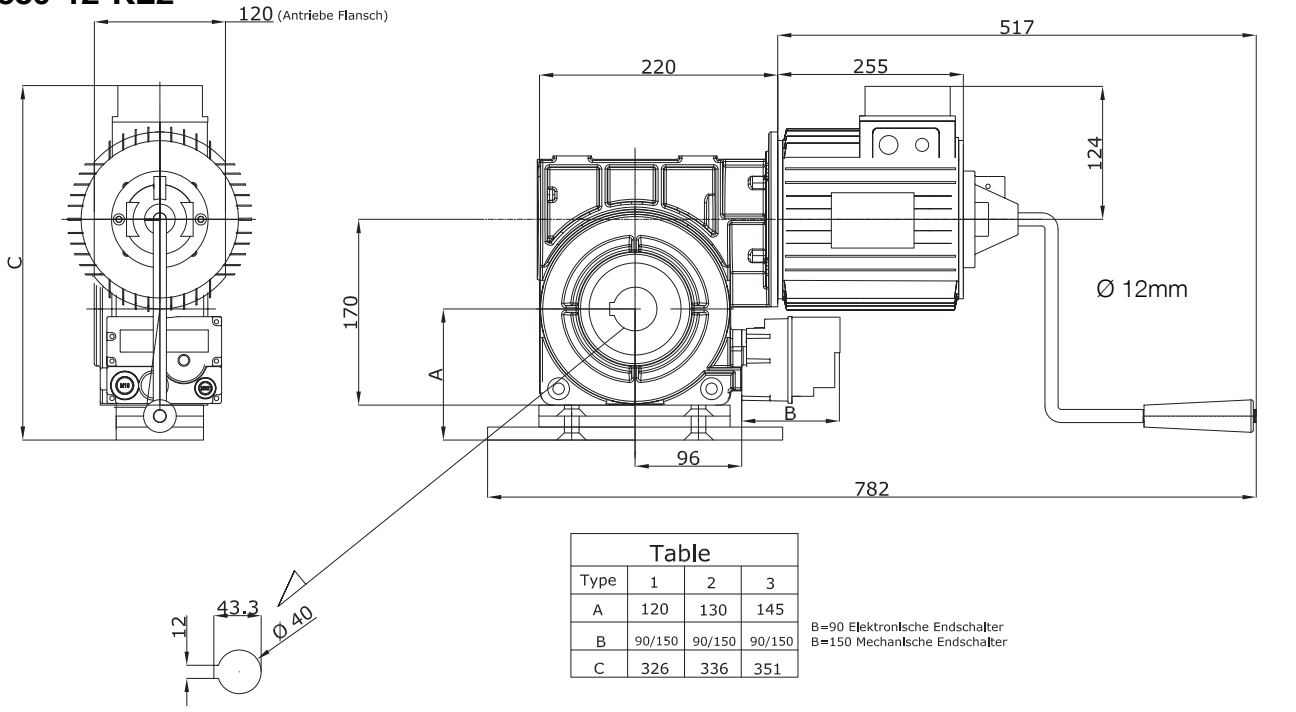
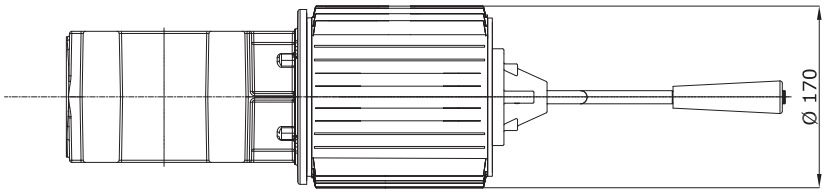


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	326	336	351

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



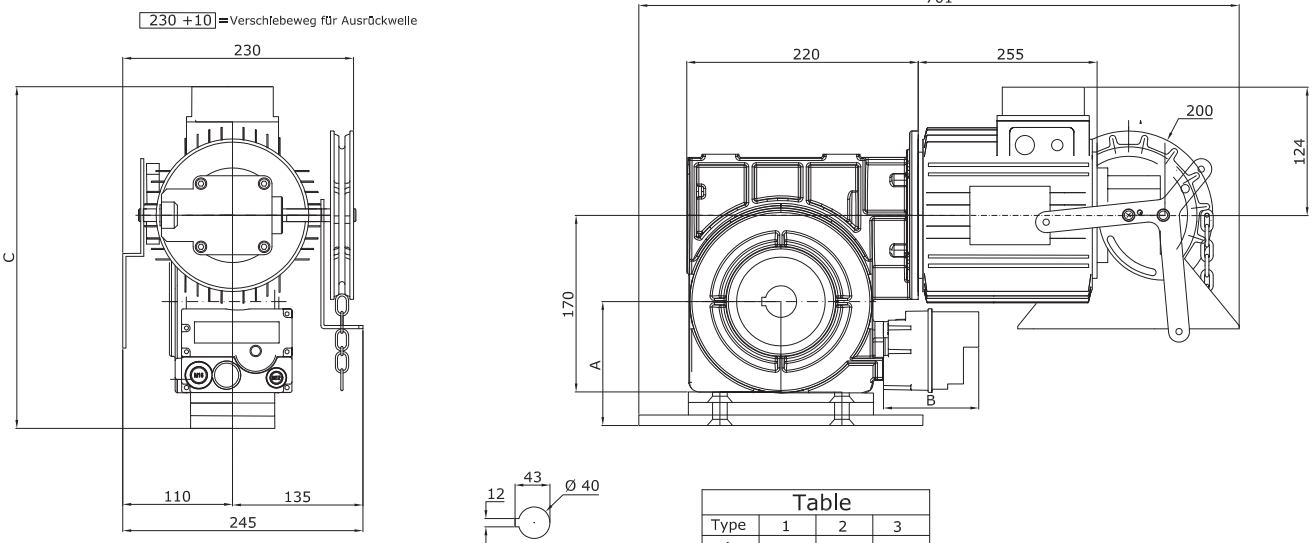
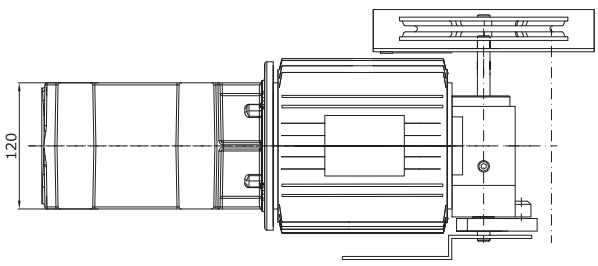


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	326	336	351

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



Gearmotors with handle and chain

RDF-750-9,5-KU
RDF-750-9,5-KE2

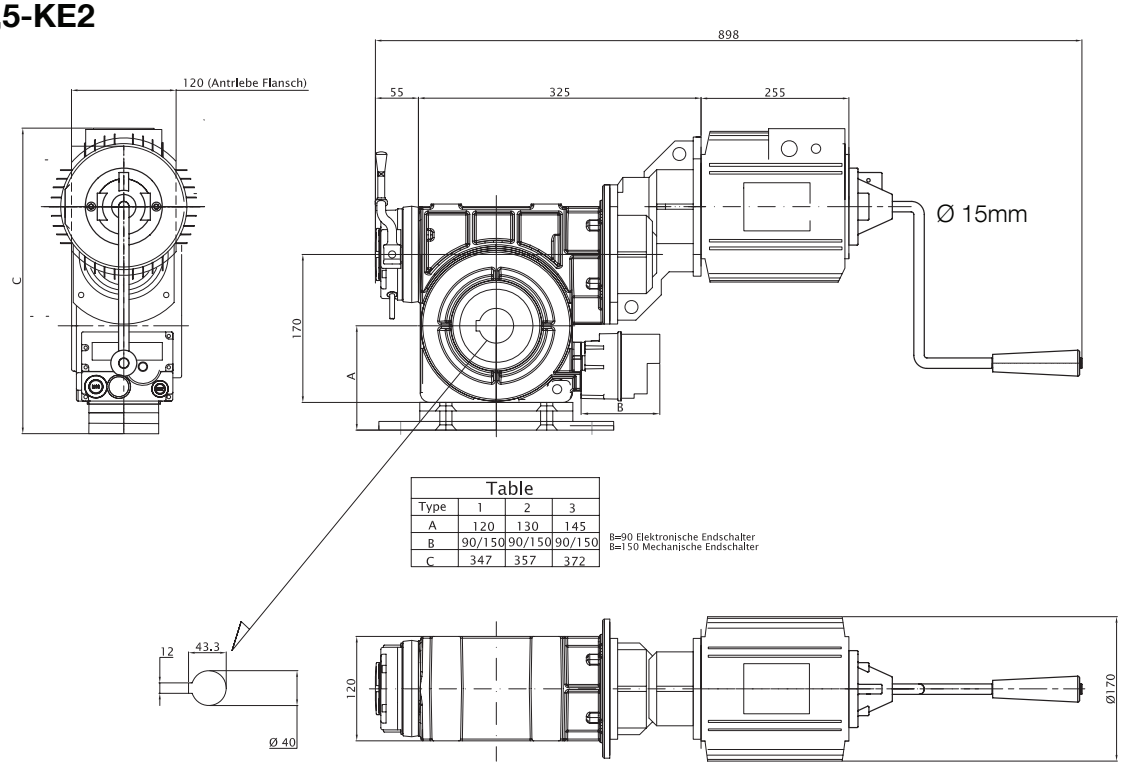


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	347	357	372

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter

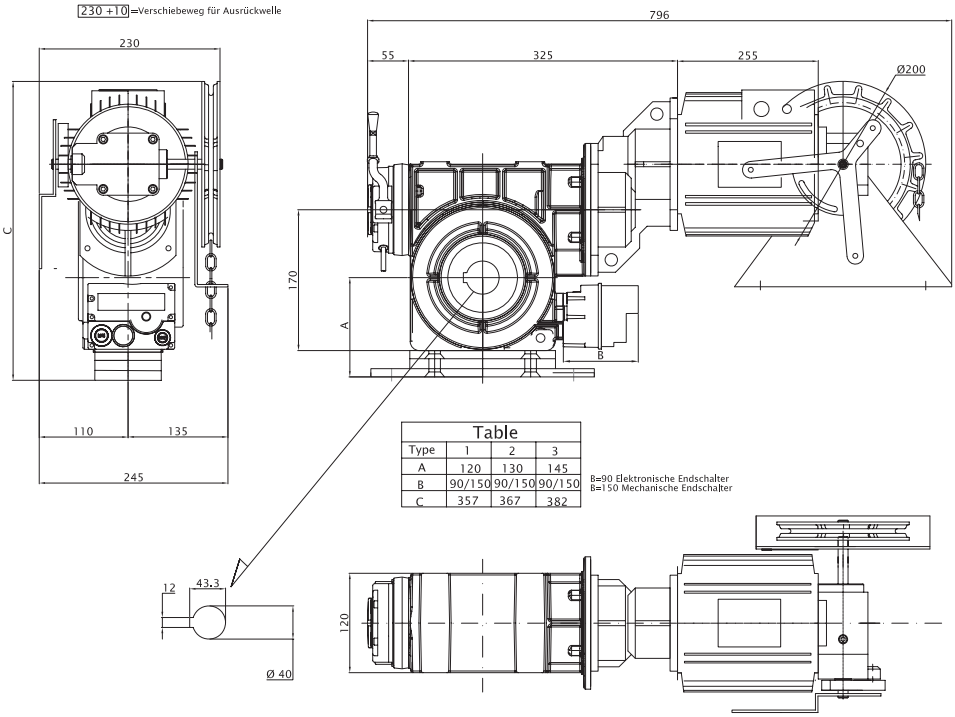
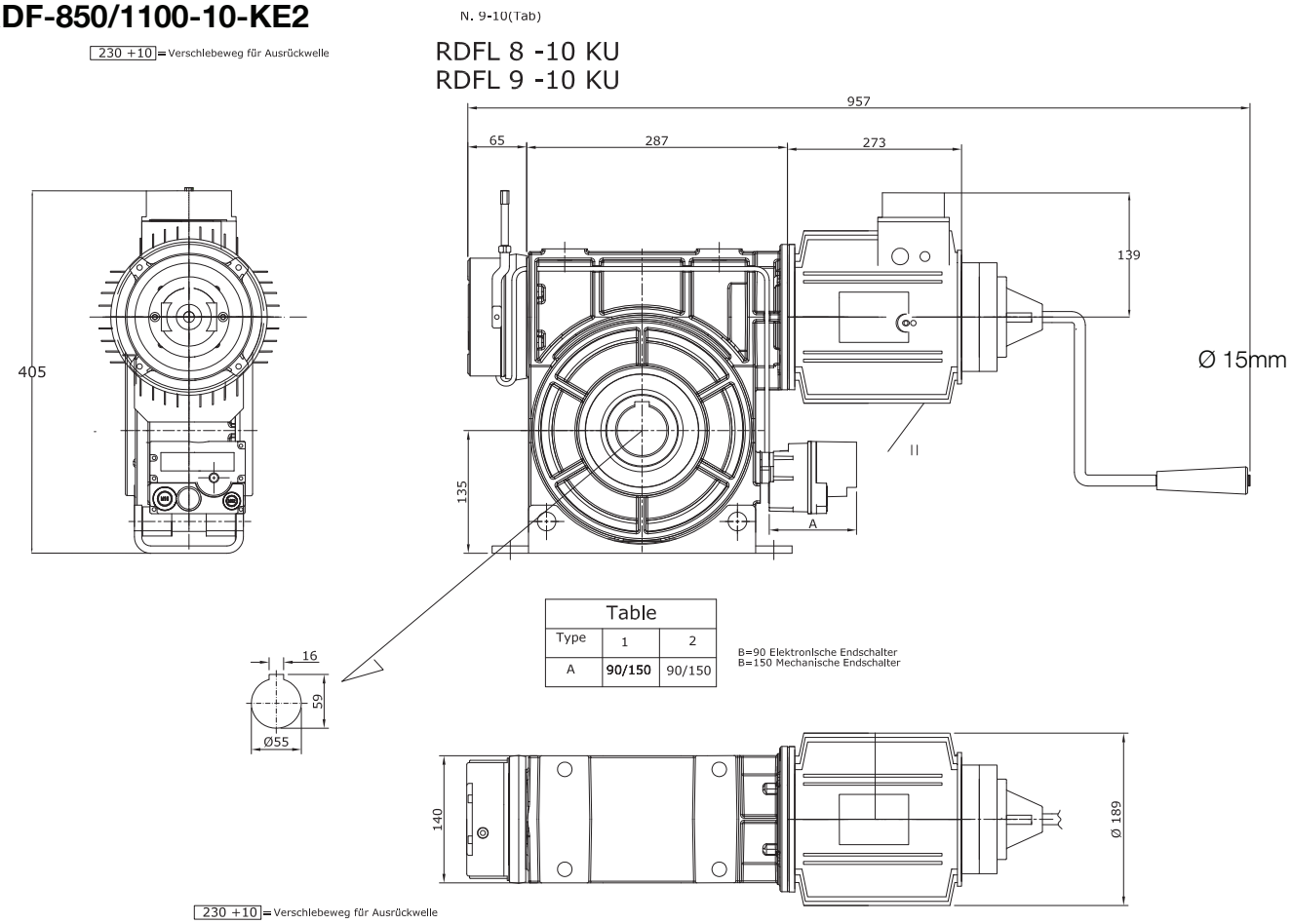


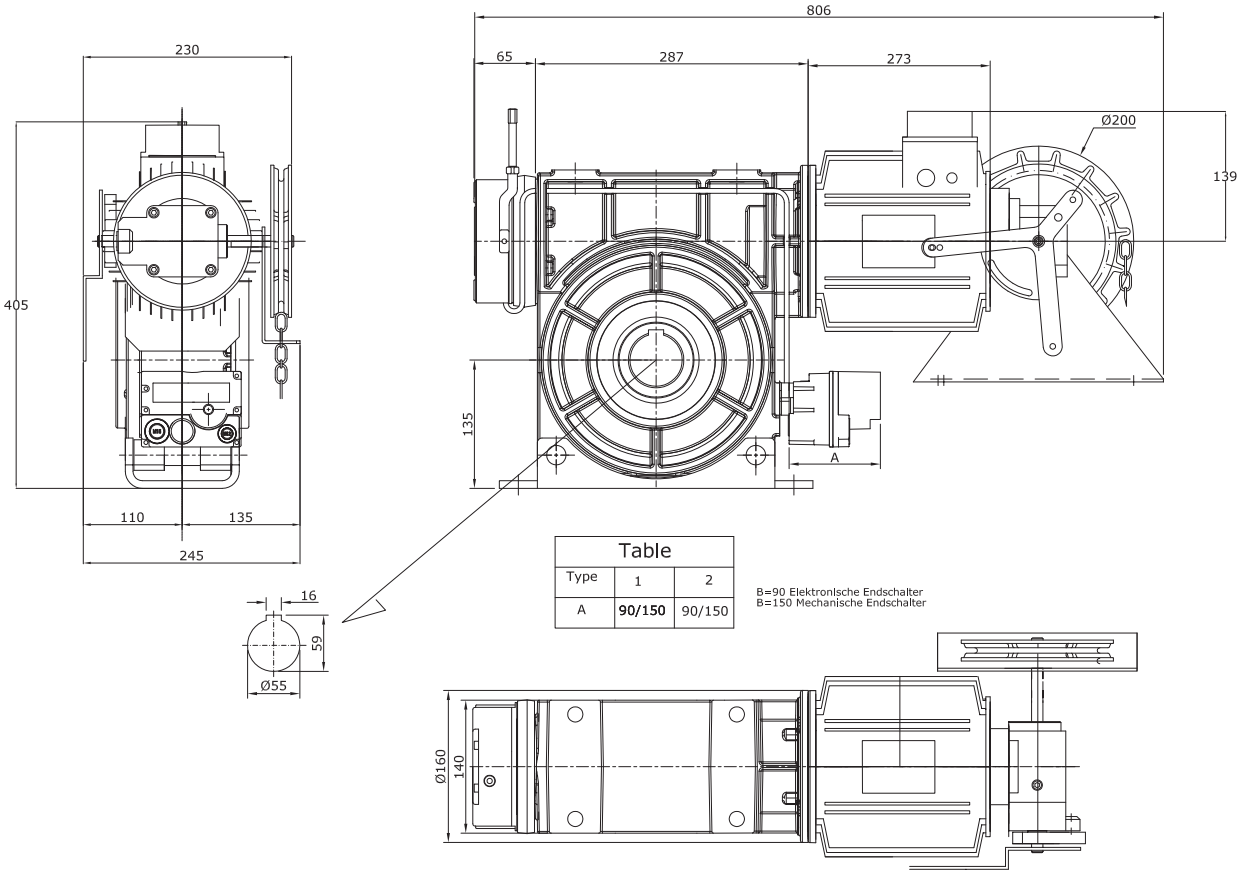
Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	357	367	382

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter

Gearmotors with handle and chain

RDF-850/1100-10-KU
RDF-850/1100-10-KE2





Gearmotors with handle and chain

RDF-1400-7-KU
RDF-1400-7-KE2

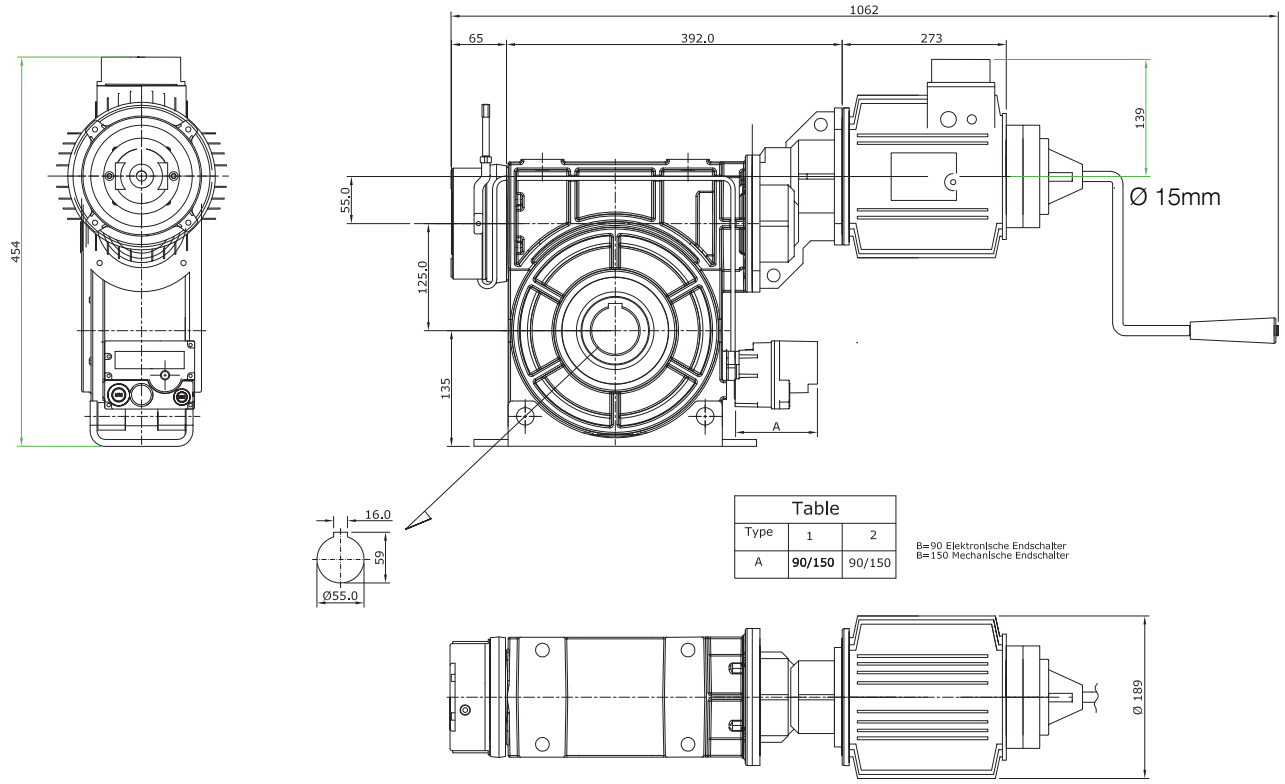


Table		
Type	1	2
A	**90/150**	90/150

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter

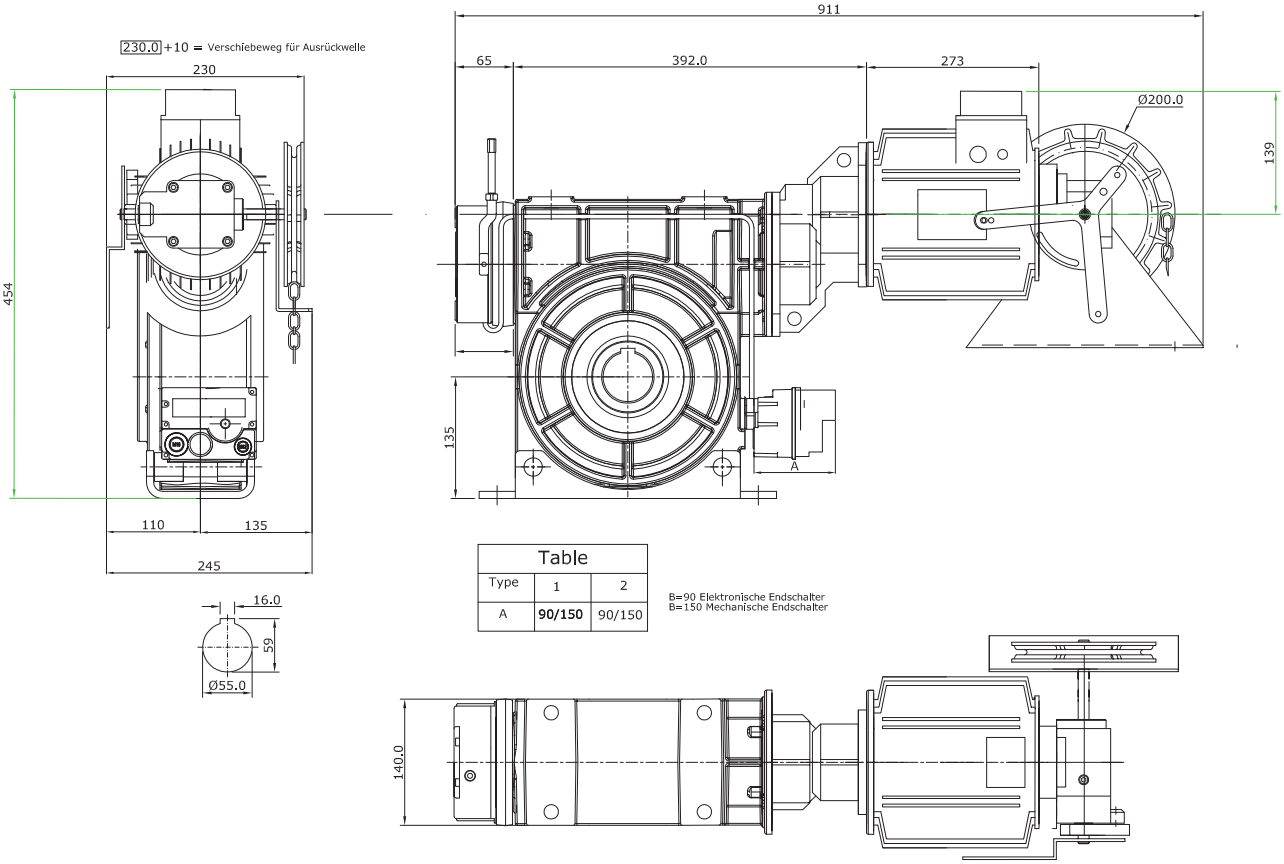


Table		
Type	1	2
A	**90/150**	90/150

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter

Gearmotors with handle and chain

RDF-250-24-KU
RDF-250-24-KE2

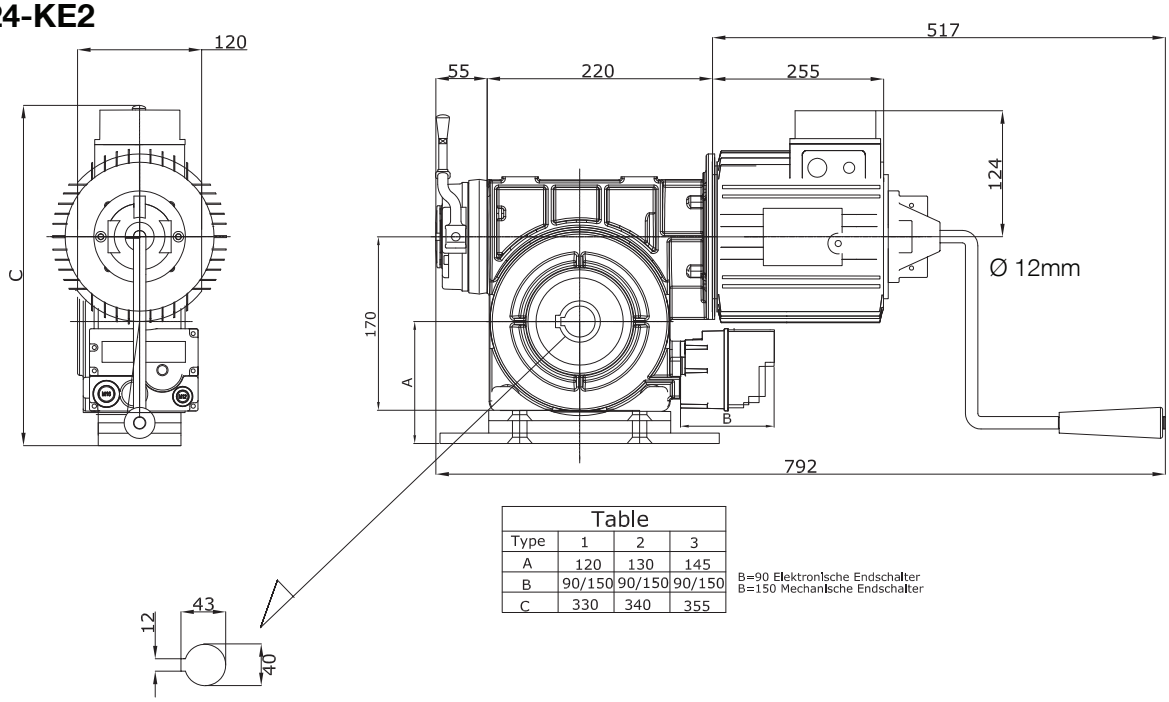
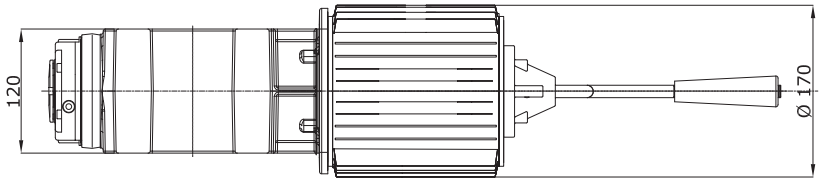


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	330	340	355

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



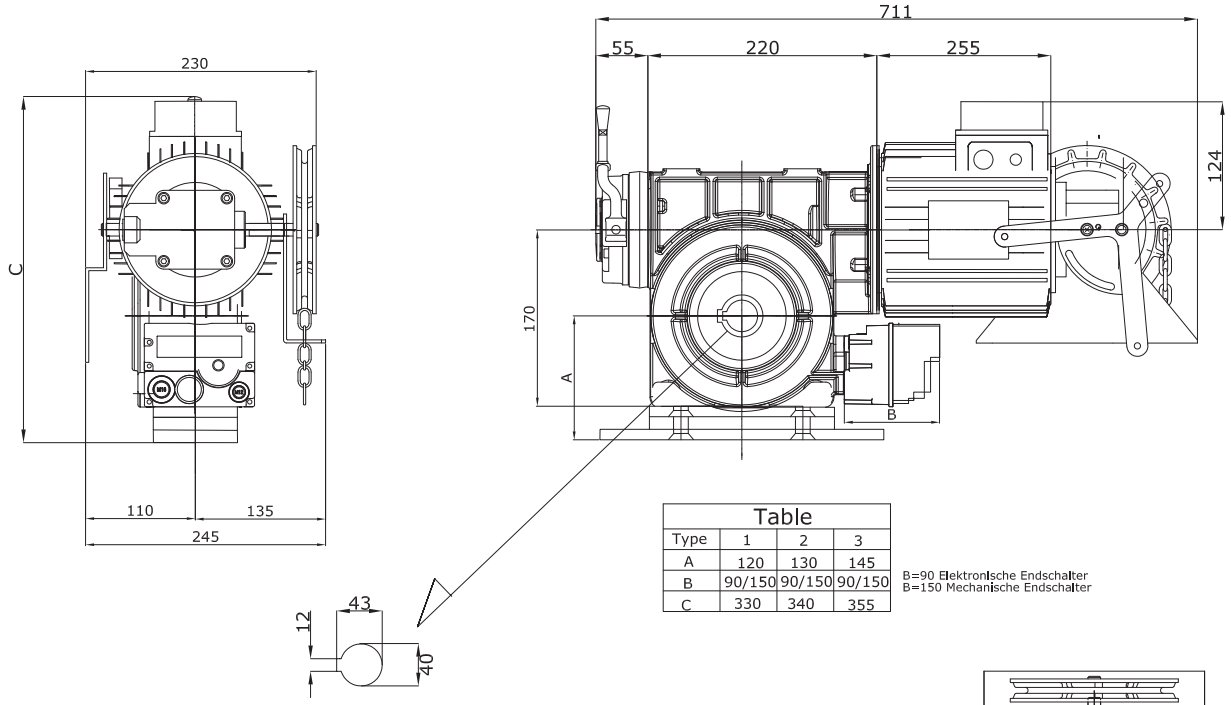
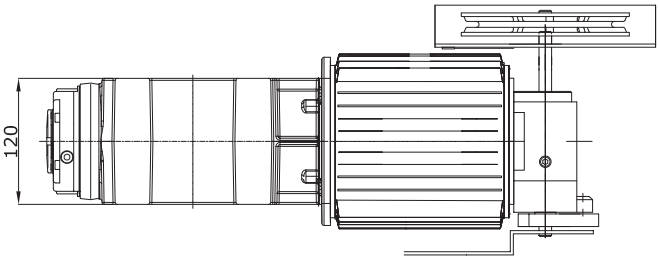


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	330	340	355

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



Gearmotors with handle and chain

RDF-350-24-KU
RDF-350-24-KE2

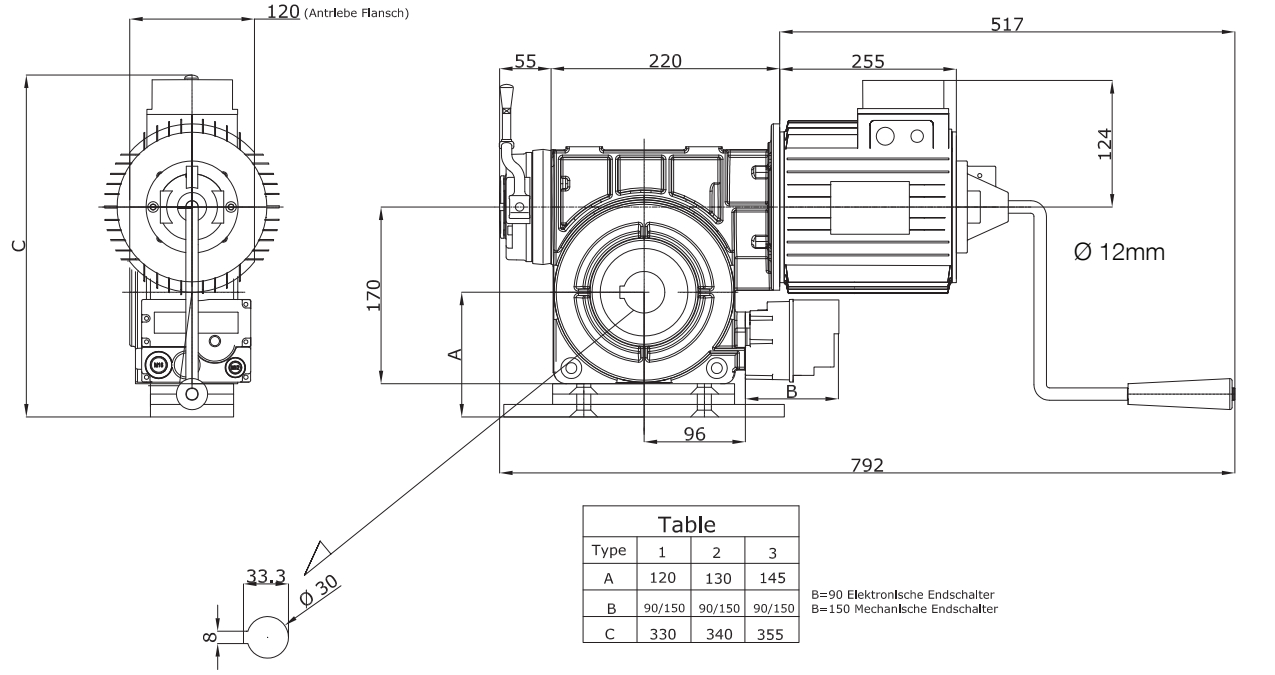
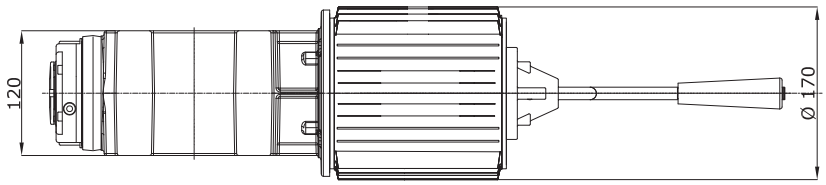


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	330	340	355

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



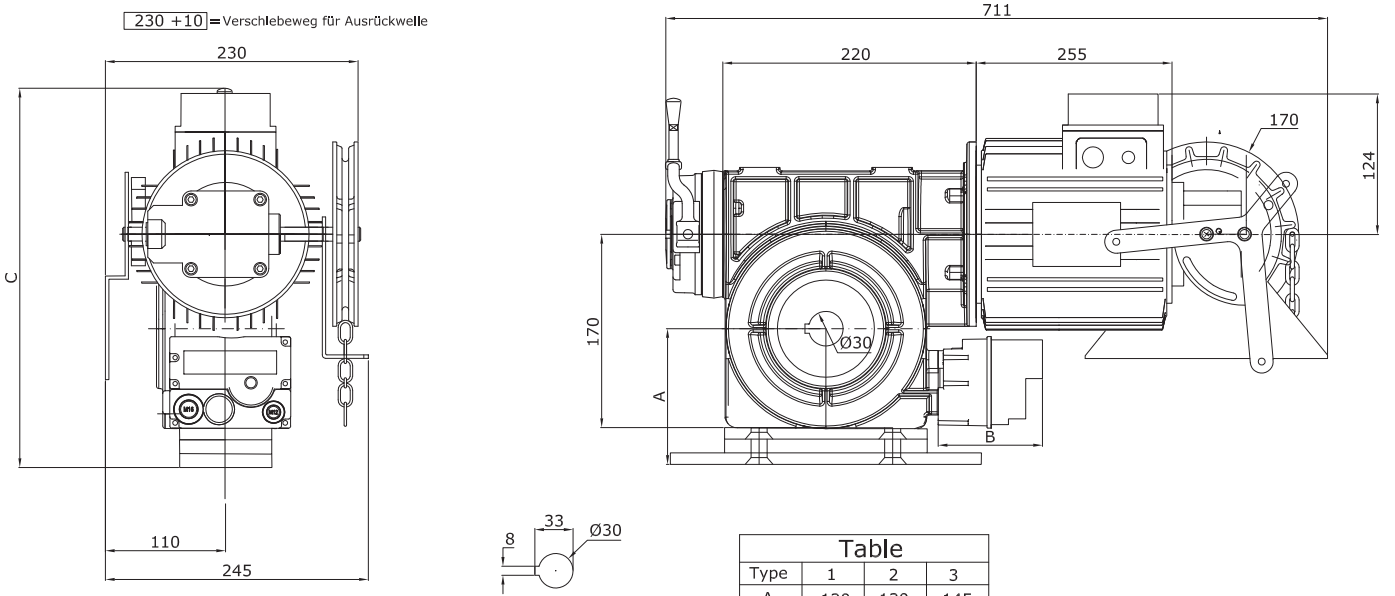
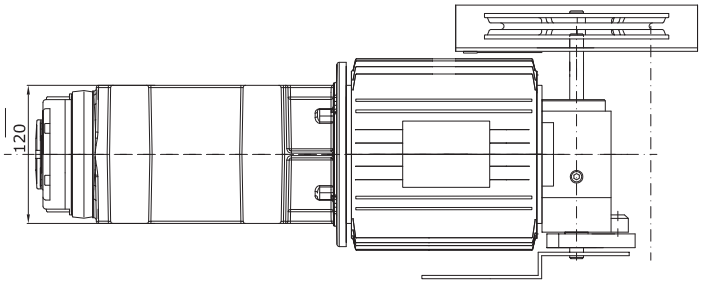


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	330	340	355

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



Gearmotors with handle and chain

RDF-500-24-KU
RDF-500-24-KE2

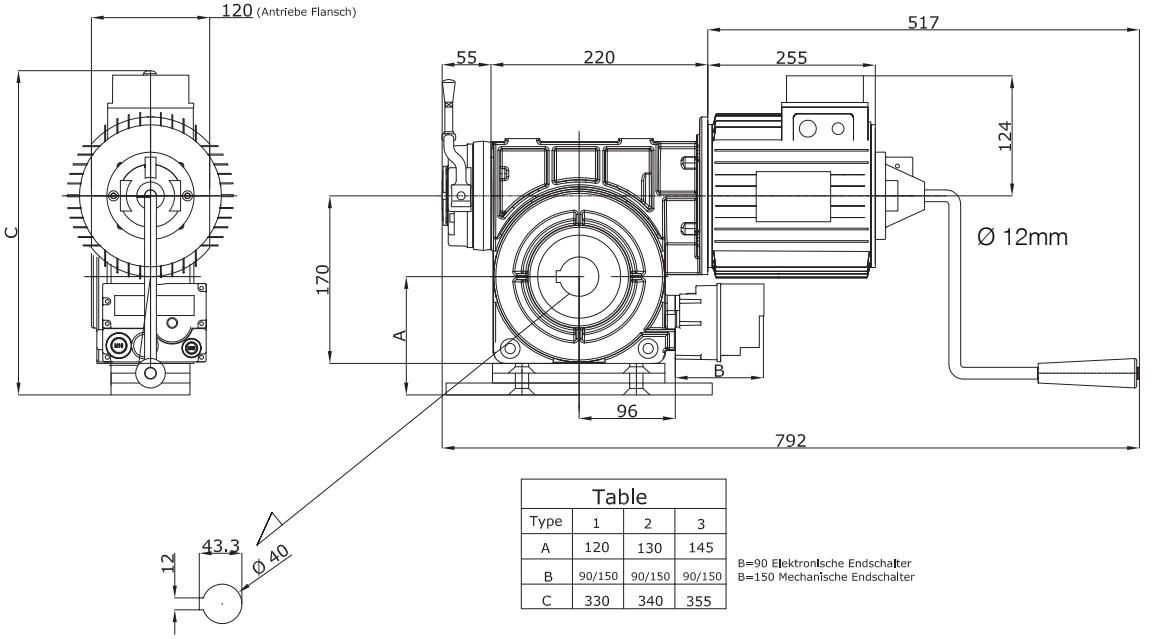
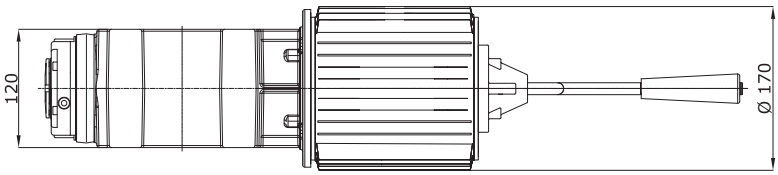


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	330	340	355

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



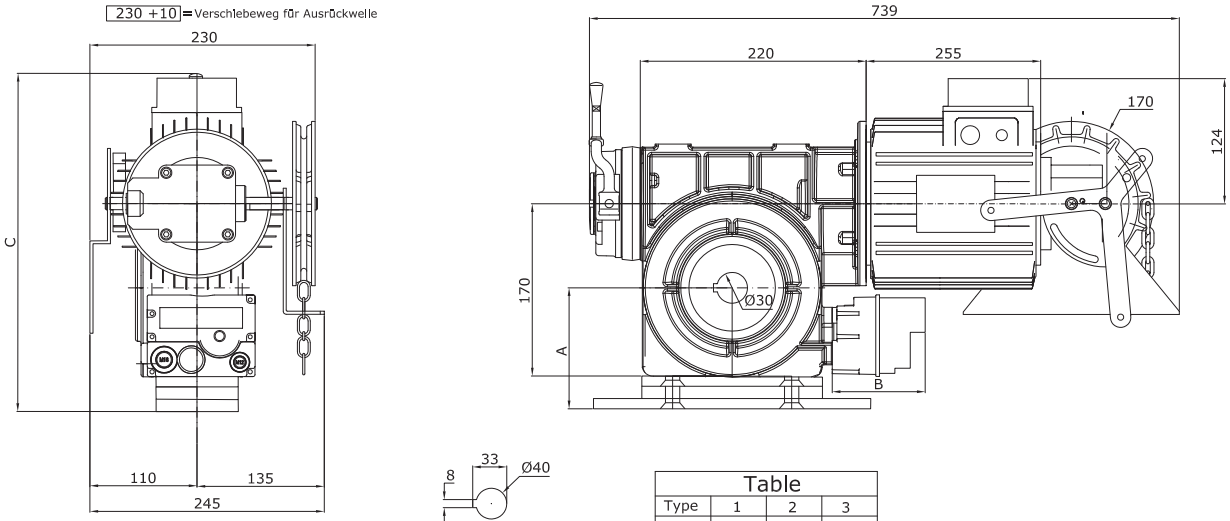
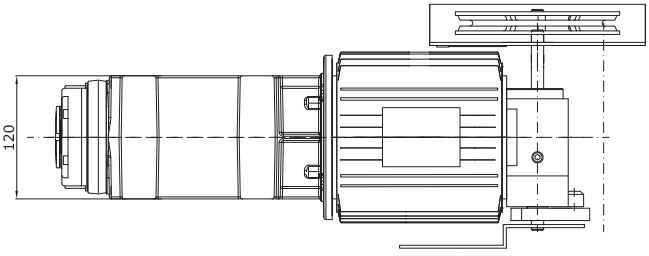


Table			
Type	1	2	3
A	120	130	145
B	90/150	90/150	90/150
C	330	340	355

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



Gearmotors with handle and chain

RDF-750-24-KU
RDF-750-24-KE2

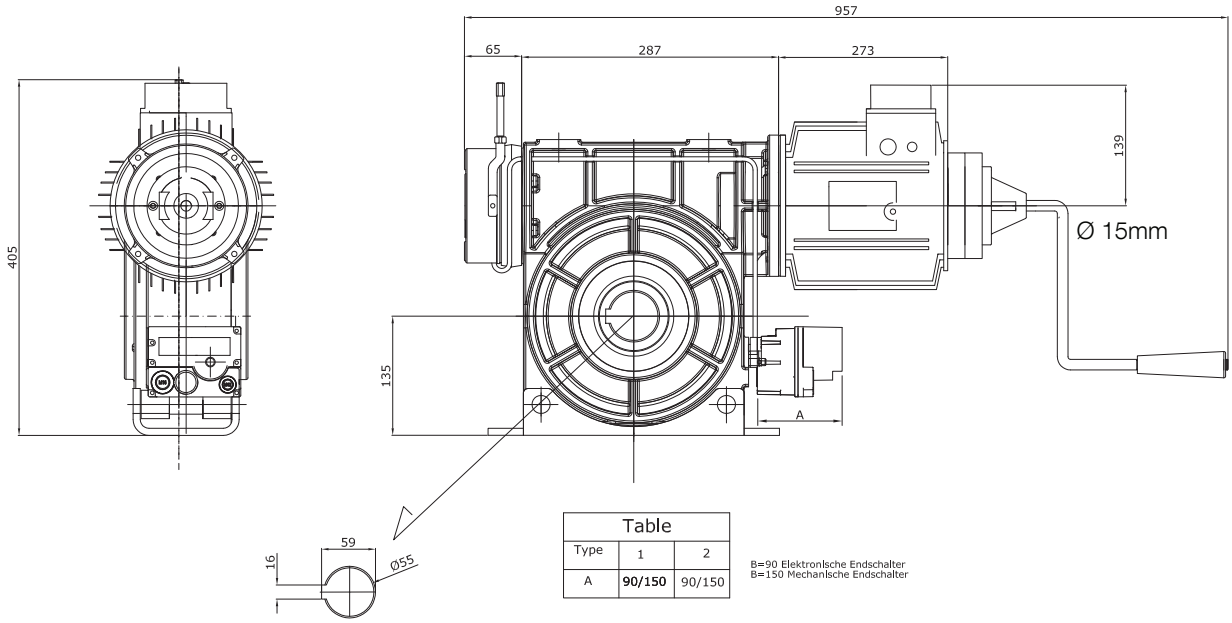
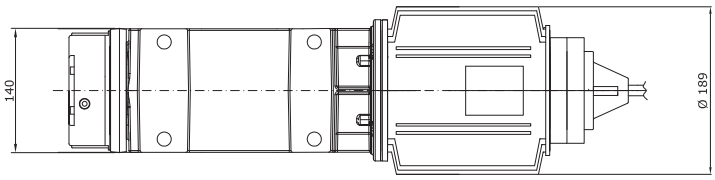


Table		
Type	1	2
A	**90/150**	90/150

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



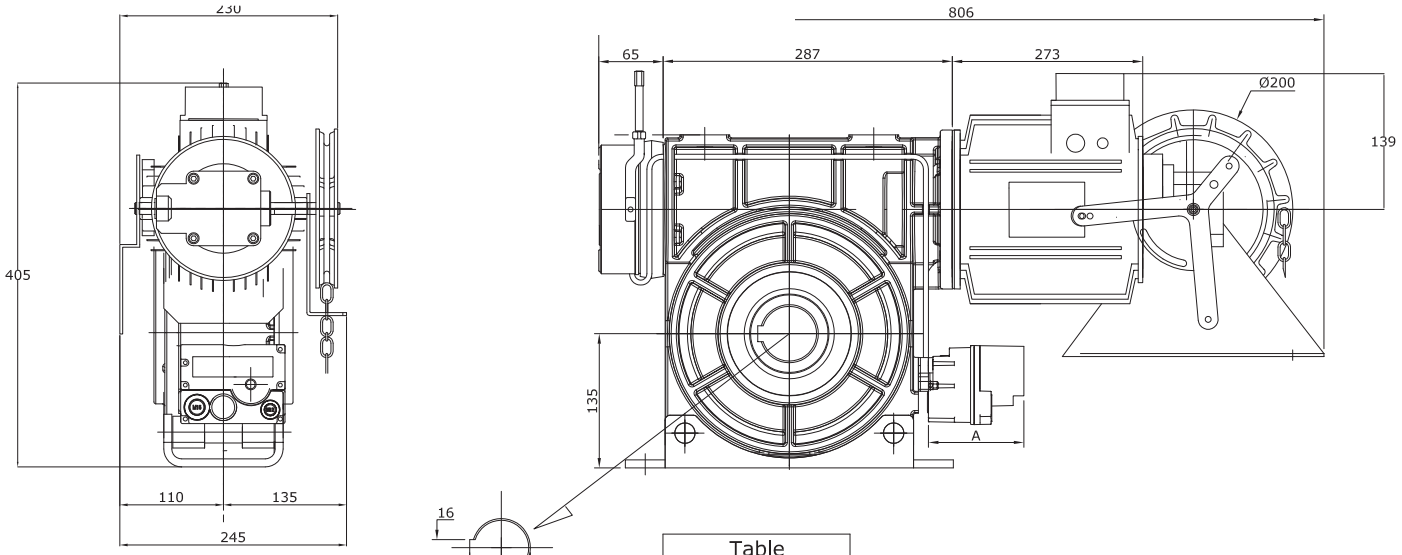
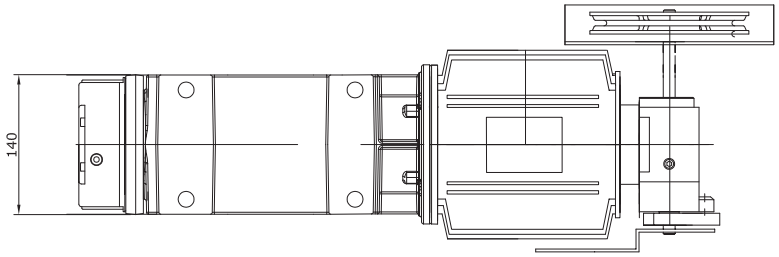


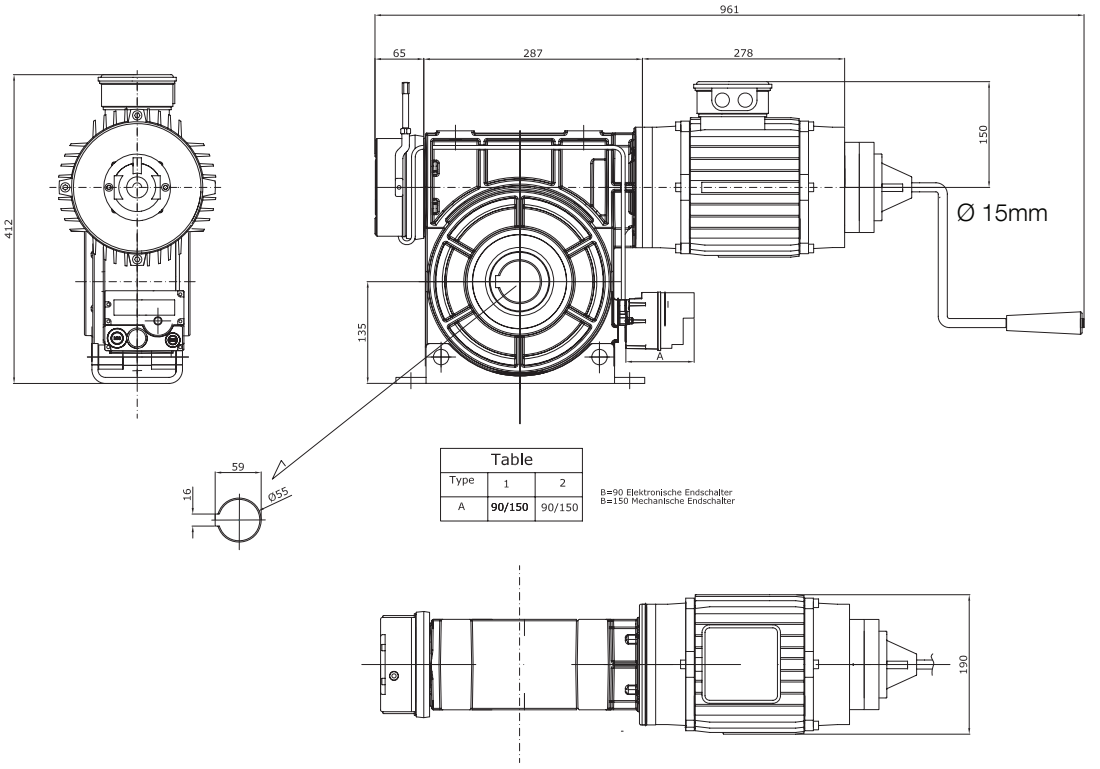
Table		
Type	1	2
A	**90/150**	90/150

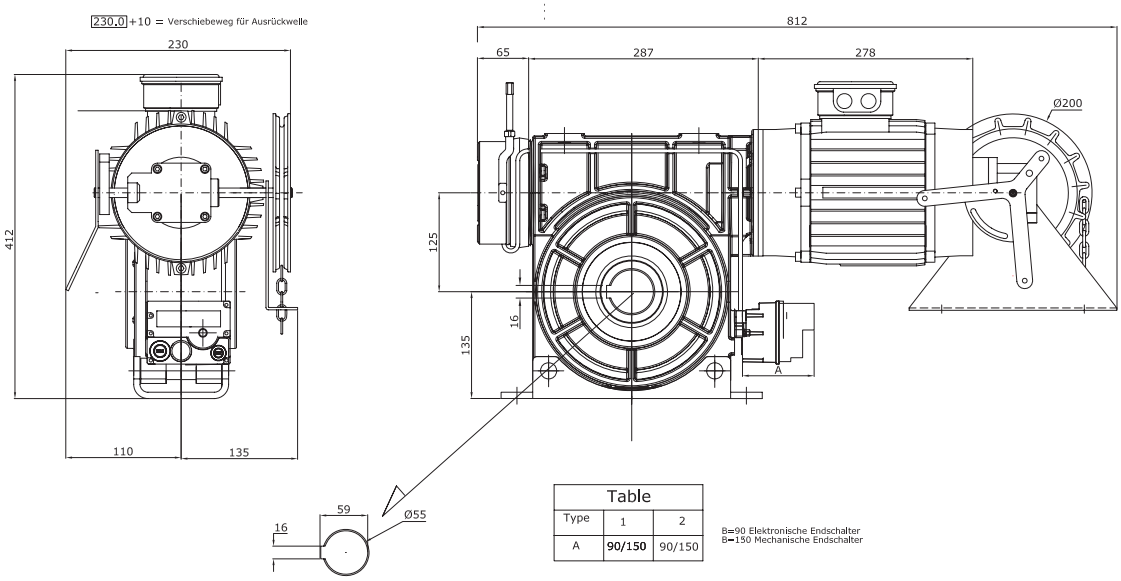
B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



Gearmotors with handle and chain

RDF-950-24-KU
RDF-950-24-KE2





Gearmotors with handle and chain

RDF-1250-24-KU
RDF-1250-24-KE2

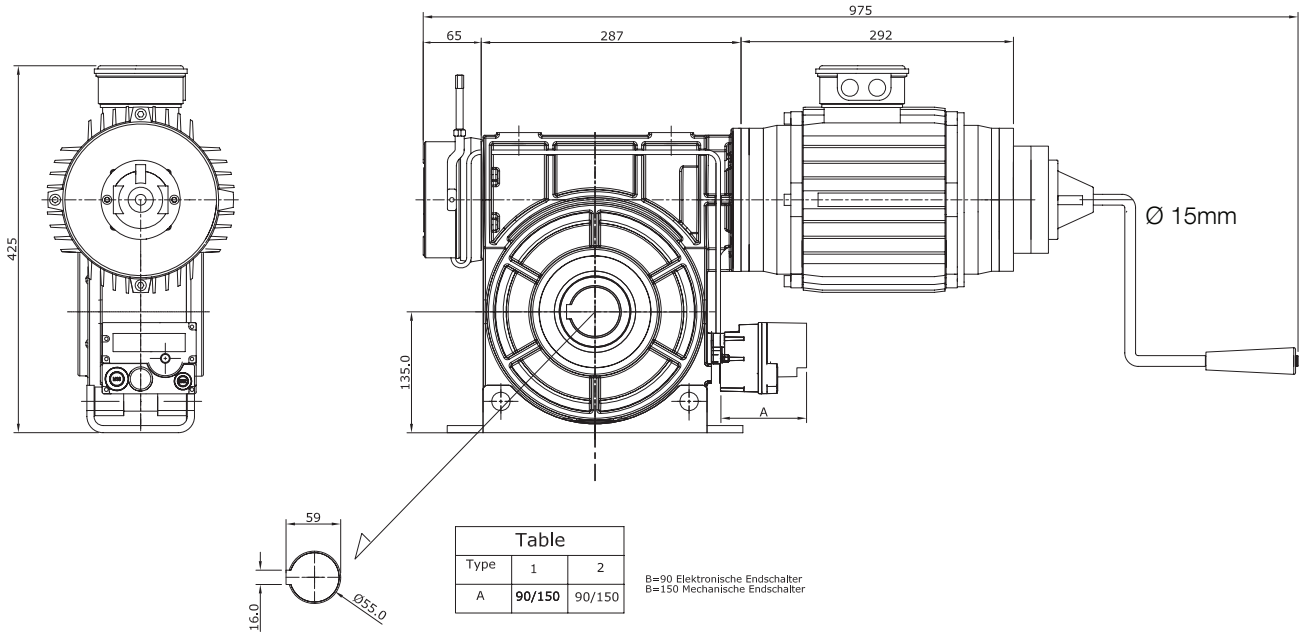
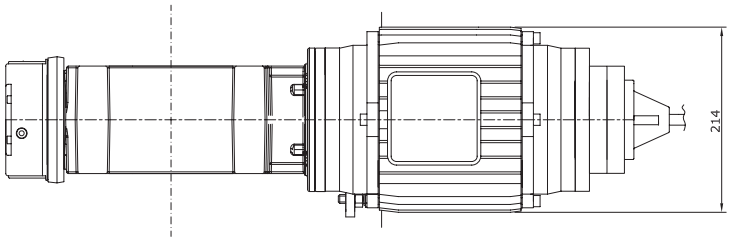


Table		
Type	1	2
A	**90/150**	90/150

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter



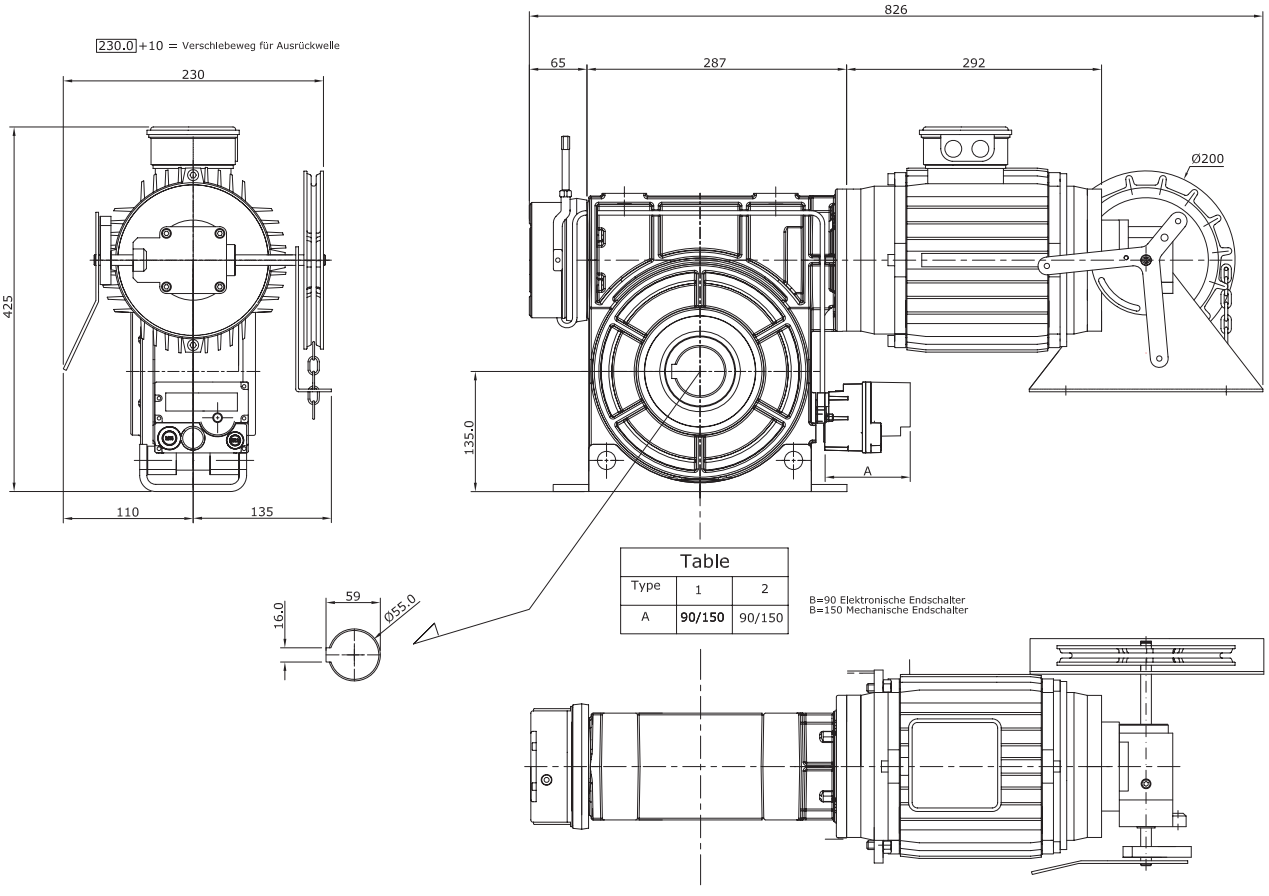


Table		
Type	1	2
A	**90/150**	90/150

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter

Gearmotors with handle and chain

RDF-1400-24-KU
RDF-1400-24-KE2

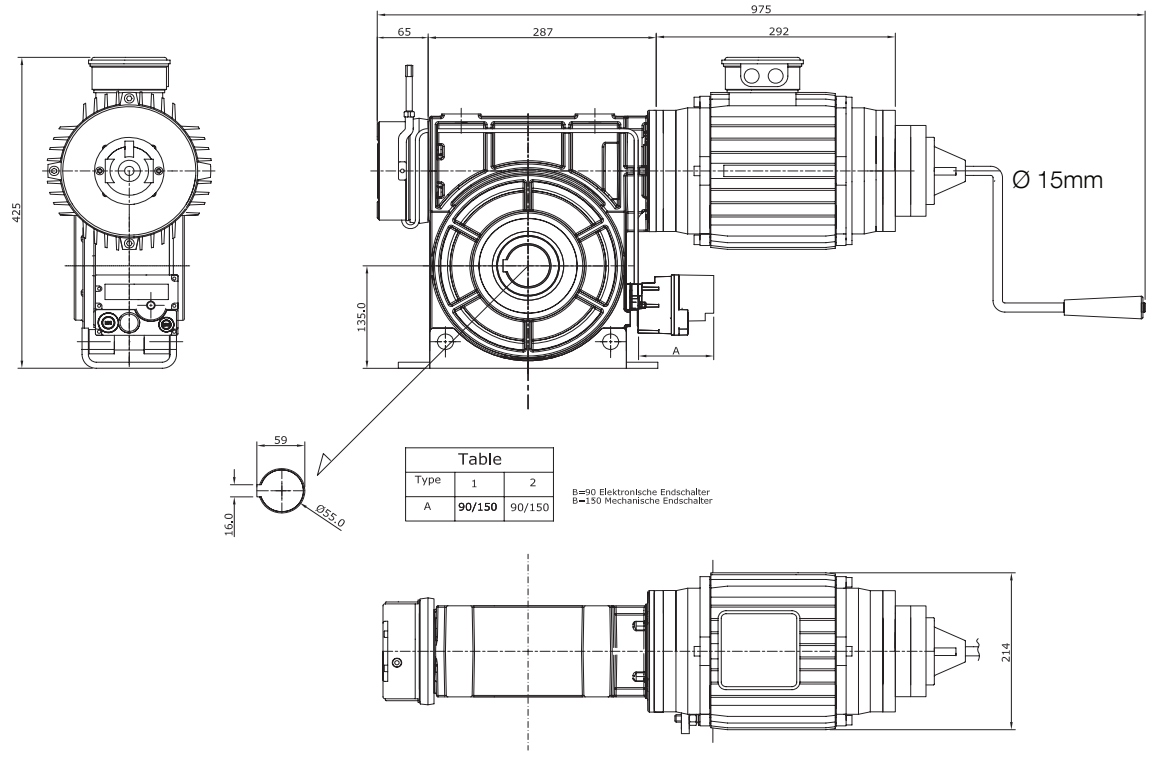


Table		
Type	1	2
A	**90/150**	90/150

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter

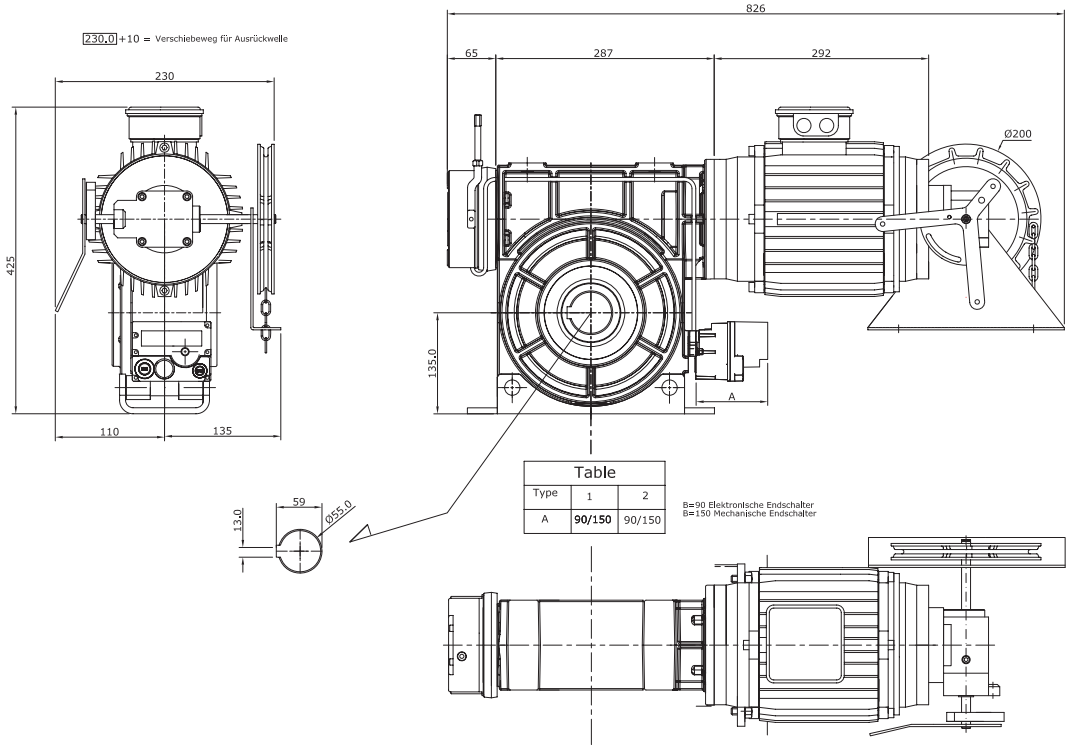


Table		
Type	1	2
A	**90/150**	90/150

B=90 Elektronische Endschalter
B=150 Mechanische Endschalter

COMBINATION OF GEARMOTORS FOR ROLLING SHUTTERS AND CONTROL STATIONS

The possible combinations of gearmotors with control stations are indicated hereunder. The gearmotors can work only with the allowed control stations.

The entire instruction manual is made up of instructions for use regarding the gearmotor and its control station. To this regard it is necessary to follow the instructions of the control station.

All gearmotors for rolling shutters can operate also with a static frequency converter. To this regard, observe the advice of the Manufacturer.

Combination Control stations/ Gearmotors for rolling shutters	140-180Nm	220-450Nm	550-1400 Nm	Motors 24RPM
UST2	X	X		
UST1	X	X		X
UST1K-2,2 / 5,5kW	X	X	X	X
UST1K-1,1kW	X			

TRANSPORT / STORAGE / DISPOSAL

The gearmotor is totally assembled and wired, ready for the connection.

To avoid any damages it is to be transported, and if necessary stored, inside the packaging provided for this purpose or another that is similar with equal characteristics.

For disposal, it is necessary to separate

– metals

– plastic elements

– electrical components

– lubricants.

TECHNICAL ASSISTANCE / SPARE PARTS / ACCESSORIES

We specifically clarify that we neither carry out tests nor are we willing to approve any spare parts and/or accessories not supplied by us.

The assembly and/or the use of products of this type could therefore compromise any predefined characteristics at the construction design level, or reflect on safety.

NICE declines all liability and disclaims any warranty whatsoever for damages, if any, caused by non-original spare parts and/or accessories.

Defects that cannot be remedied internally are to be eliminated only by the Door manufacturer or another specialised Company, from which it is possible to also request any necessary spare parts.